

On complete reducibility in characteristic p

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with an appendix by Zhiwei Yun

in memoriam Vikram Mehta

Abstract

Let G be a reductive group over a field k which is algebraically closed of characteristic $p \neq 0$. We prove a structure theorem for a class of subgroup schemes of G when p is bounded below by the Coxeter number of G . As applications, we derive semisimplicity results generalizing earlier results of Serre proven in 1998 and also obtain an analogue of Luna's étale slice theorem for suitable bounds on p .

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French metadata in the source. *Titre.* Complète réductibilité en caractéristique p . *Résumé.* Soit G un groupe réductif sur un corps k qui est algébriquement clos de caractéristique $p \neq 0$. Nous prouvons un théorème de structure pour une classe de schémas en sous-groupes de G lorsque p est minoré par le nombre de Coxeter de G . En guise d'applications, nous en déduisons des résultats de semisimplicité qui généralisent des résultats antérieurs de Serre établis en 1998 et nous obtenons également un analogue du théorème de Luna, *étale slice*, pour des bornes convenables sur p .

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1 Introduction

We work over an algebraically closed field k of characteristic $p > 0$.

In [S1], Serre showed that if semisimple representations V_i of a group Γ are such that

$$\sum_i (\dim V_i - 1) < p,$$

then their tensor product is semisimple. In [S2], he more generally considers the case where Γ is a subgroup of $G(k)$, for G a reductive group, and where Γ is G -cr, meaning that whenever Γ is contained in a parabolic subgroup P , it is already contained in a Levi subgroup of P . For

$$G = \prod_i \mathrm{GL}(V_i),$$

this is equivalent to the semisimplicity of the representations V_i of Γ . For a representation V of G , one can then ask under what conditions V becomes semisimple when considered as a representation of Γ . In [S2, Theorem 6, page 25], Serre shows that this is the case when the Dynkin height $\mathrm{ht}_G(V)$ is less than p . For

$$G = \prod_i \mathrm{GL}(V_i), \quad V = \bigotimes_i V_i,$$

one has

$$\mathrm{ht}_G(V) = \sum_i (\dim V_i - 1).$$

In [D], the results of [S1] were generalized to the case when the V_i are semisimple representations of a group scheme $\mathcal{G}$. In this paper, we consider the case when $\mathcal{G}$ is a subgroup scheme of a reductive group G , and generalize [S2], see 4.11, and [D], see 2.5. As in [D], we first have to prove a structure theorem, 2.5, on doubly saturated, see 2.4, subgroup schemes $\mathcal{G}$ of reductive groups G . The proof makes crucial use of a result of Zhiwei Yun on root systems. The appendix contains the result.

In Section 5, we consider a reductive group acting on an affine variety X and a point x of X whose orbit $G \cdot x$ is closed in X . We prove a schematic analogue of [BR, Propositions 7.4, 7.6] under some conditions on the characteristic of k . More precisely, if X embeds in a G -module V of low height, then we obtain, as a consequence of 2.5(2), an analogue of Luna's étale slice theorem, 5.1 and 5.6. In [BR] the language of schemes was not used, and as a consequence the orbit $G \cdot x$ had to be assumed "separable". An orbit $G \cdot x$ is separable if and only if the stabilizer G_x is reduced.

2 Saturation and Infinitesimal saturation

2.1

Let G be a reductive algebraic group over k . Our terminology is that of [SGA3]: reductive implies smooth and connected. By an algebraic group we will mean an affine group scheme of finite type over k . Fix a maximal torus T_G and a Borel subgroup containing T_G , which determines a root system R and a set of positive roots R^+ . Let $\langle, \rangle$ be the natural pairing between the characters and cocharacters and, for each root α , let $\alpha^\vee$ be the corresponding coroot.

If the root system R associated to G is irreducible, the Coxeter number h_G of R and of G admits the following equivalent descriptions:

- (1) It is the order of the Coxeter elements of the Weyl group W . This shows that R and the dual root system $R^\vee$ have the same Coxeter number.
- (2) Let α_0 be the highest root and $\sum n_i \alpha_i$ its expression as a linear combination of the simple roots. One has

$$h_G = 1 + \sum_i n_i. \tag{2.1.1}$$

- (3) Applying this to the dual root system, one gets

$$h_G = \langle \rho, \beta^\vee \rangle + 1,$$

where ρ is half the sum of positive roots and $\beta^\vee$ is the highest coroot. Indeed, ρ is also the sum of the fundamental weights ω_i , and

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}.$$

For a general reductive group G , define the Coxeter number h_G to be the largest among the ones for the irreducible components of R .

It follows from (2) above that if G is a reductive group, U the unipotent radical of a Borel subgroup, and $\mathfrak{u} := \text{Lie}(U)$, the descending central series of $\mathfrak{u}$, defined by $Z^1\mathfrak{u} = \mathfrak{u}$ and

$$Z^i\mathfrak{u} = [\mathfrak{u}, Z^{i-1}\mathfrak{u}],$$

satisfies

$$Z^{h_G}(\mathfrak{u}) = 0. \quad (2.1.2)$$

For the group U , similarly $Z^{h_G}(U) = (1)$.

The Lie algebra $\mathfrak{g}$ of G is a p -Lie algebra. Let $X \mapsto X^{[p]}$ denote the p -power operation on $\mathfrak{g}$. If $\mathfrak{n} \in \mathfrak{g}$ is nilpotent and if $p \geq h_G$, then $\mathfrak{n}^{[p]} = 0$. To check this, we may assume that $\mathfrak{n}$ is in the Lie algebra $\mathfrak{u}$ of the unipotent radical U of a Borel subgroup. For each positive root α , let X_α be a basis for the root subspace $\mathfrak{g}_\alpha$. Express

$$\mathfrak{n} = \sum_{\alpha \in R^+} a_\alpha X_\alpha.$$

Each X_α is the infinitesimal generator of an additive group. It follows that $X_\alpha^{[p]} = 0$ for each α . Observe that

$$\mathfrak{n}^{[p]} = \sum_{\alpha \in R^+} a_\alpha^p X_\alpha^{[p]} \pmod{Z^p\mathfrak{u}}, \quad (2.1.3)$$

and $Z^p\mathfrak{u} = 0$, as $p \geq h_G$, see [Mc1, page 10].

2.2

Let $\mathfrak{g}_{\text{nilp}}$, respectively G^{u} , be the reduced subscheme of $\text{Lie}(G)$, respectively G , with points the nilpotent, respectively unipotent, elements. Let U be the unipotent radical of a Borel subgroup. For $p \geq h_G$, the Campbell–Hausdorff group law $\circ$ makes sense in characteristic p and turns $\mathfrak{u} := \text{Lie}(U)$ into an algebraic group over k . This is so since $Z^p\mathfrak{u} = 0$. Further, there is a unique isomorphism

$$\exp : (\text{Lie}(U), \circ) \xrightarrow{\sim} U \quad (2.2.1)$$

equivariant for the action of B and whose differential at the origin is the identity. If, in addition, the simply connected covering of the derived group of G is an étale covering, which is the case for $p > h_G$ and could fail when $p = h_G$ because of the presence of $\text{SL}(p)$ as factors in the covering, then there is a unique G -equivariant isomorphism, see [S2, Theorem 3, page 21] and Section 6 for details,

$$\exp : \mathfrak{g}_{\text{nilp}} \longrightarrow G^{\text{u}}, \quad (2.2.2)$$

which induces (2.2.1) on each unipotent radical of a Borel subgroup. Let

$$\log : G^{\text{u}} \longrightarrow \mathfrak{g}_{\text{nilp}}$$

denote its inverse.

For u a unipotent element of $G(k)$, one defines the t -power map $t \mapsto u^t$, from $\mathbf{G}_a$ to G , by

$$t \longmapsto \exp(t \log u). \quad (2.2.3)$$

For $G = \mathrm{GL}(V)$ such a map $t \mapsto u^t$ is more generally defined for any u in G such that $u^p = 1$. It is given by the truncated binomial expression [S1, 4.1.1, page 524]:

$$t \mapsto u^t := \sum_{i < p} \binom{t}{i} (u - 1)^i. \quad (2.2.4)$$

Similarly, $X \in \mathrm{End}(V)$ such that $X^{[p]} = 0$ defines a morphism $t \mapsto \exp(tX)$ from $\mathbf{G}_a$ to $\mathrm{GL}(V)$ given by the truncated exponential series

$$t \mapsto I + tX + \frac{(tX)^2}{2!} + \cdots + \frac{(tX)^{p-1}}{(p-1)!}. \quad (2.2.5)$$

Until the end of Section 3, we make the following assumptions on the reductive group G .

Assumption 2.3. *Let $\tilde{G}$ be the simply connected covering of the derived group G' . We assume that $p \geq h_G$ and that the map $\tilde{G} \rightarrow G'$ is étale.*

In particular, by 2.1 and 2.2, the exponential map (2.2.2) is defined, every unipotent element in $G(k)$ is of order p , and every nilpotent in $\mathrm{Lie}(G)$ is p -nilpotent. One can then define the notions of saturation and infinitesimal saturation of subgroup schemes $\mathcal{G} \subset G$ as follows; see Remark 2.19 for the case when $G = \mathrm{PGL}(p)$.

Definition 2.4 (S1, §4, D, Definition 1.5). *(1) A subgroup scheme $\mathcal{G} \subset G$ is called saturated if, for every $u \in \mathcal{G}(k)$ which is unipotent, the homomorphism $t \mapsto u^t$, (2.2.3), from $\mathbf{G}_a$ to G factors through $\mathcal{G}$.*

(2) A subgroup scheme $\mathcal{G} \subset G$ is called infinitesimally saturated if, for every nilpotent $X \in \mathrm{Lie}(\mathcal{G})$, the morphism $t \mapsto \exp(tX)$, (2.2.2), from $\mathbf{G}_a$ to G factors through $\mathcal{G}$.

(3) $\mathcal{G}$ is doubly saturated if it is saturated and infinitesimally saturated.

An element of $\mathrm{Lie}(\mathcal{G})$ is nilpotent if and only if it is nilpotent as an element of $\mathrm{Lie}(G)$. The reference to the exponential map (2.2.2) in (2) therefore makes sense. One way to see that the notions of nilpotence for elements of $\mathrm{Lie}(\mathcal{G})$ and $\mathrm{Lie}(G)$ coincide is to observe that the inclusion of $\mathrm{Lie}(\mathcal{G})$ in $\mathrm{Lie}(G)$ is a morphism of p -Lie algebras and that X is nilpotent if and only if it is killed by an iterated p -power map, i.e.

$$X^{[p^\ell]} = 0.$$

Let $\mathcal{G}^0$ be the identity component of $\mathcal{G}$ and $\mathcal{G}_{\mathrm{red}}^0$ the reduced subscheme of $\mathcal{G}^0$.

Theorem 2.5. *Let $\mathcal{G} \subset G$ be a k -subgroup scheme which is infinitesimally saturated. Assume that, if $p = h_G$, $\mathcal{G}_{\mathrm{red}}^0$ is reductive. Then:*

(1) The group $\mathcal{G}_{\mathrm{red}}^0$ and its unipotent radical $R_u(\mathcal{G}_{\mathrm{red}}^0)$ are normal subgroup schemes of $\mathcal{G}$, and the quotient group scheme

$$\mathcal{G}^0 / \mathcal{G}_{\mathrm{red}}^0$$

is of multiplicative type.

(2) If $\mathcal{G}_{\mathrm{red}}^0$ is reductive, there exists a central, connected subgroup scheme of multiplicative type $M \subset \mathcal{G}^0$ such that the morphism

$$M \times \mathcal{G}_{\mathrm{red}}^0 \longrightarrow \mathcal{G}^0$$

realizes $\mathcal{G}^0$ as a quotient of $M \times \mathcal{G}_{\mathrm{red}}^0$.

We note that since k is assumed to be algebraically closed, a group scheme of multiplicative type is simply a diagonalizable group scheme.

The proof of part (2) of 2.5 will occupy most of Section 2, until 2.16. Part (1) will be proven in Section 3.

By [D, Lemma 2.3], the conclusions of 2.5 hold for $\mathcal{G}$ if and only if they hold for the identity component $\mathcal{G}^0$. Until the end of Section 3, we will assume that $\mathcal{G}$ is connected.

Lemma 2.6. *If $\mathcal{G}$ is an infinitesimally saturated subgroup scheme of G , every nilpotent element $\mathfrak{n}$ of $\mathrm{Lie}(\mathcal{G})$ is in $\mathrm{Lie}(\mathcal{G}_{\mathrm{red}})$.*

Proof. As $\mathbf{G}_a$ is reduced, the morphism $t \mapsto \exp(t\mathfrak{n})$ maps $\mathbf{G}_a$ to $\mathcal{G}_{\mathrm{red}} \subset \mathcal{G}$. Identifying $\mathrm{Lie}(\mathbf{G}_a)$ with a copy of the ground field, the image of 1 in $\mathrm{Lie}(\mathbf{G}_a)$ is $\mathfrak{n}$. $\square$

Part (2) of 2.5 is a corollary of 2.6 and of the following theorem, which does not refer to G anymore.

Theorem 2.7. *Let $\mathcal{G}$ be a connected algebraic group such that*

- (a) $\mathcal{G}_{\mathrm{red}}$ is reductive,
- (b) any nilpotent element of $\mathrm{Lie}(\mathcal{G})$ is in $\mathrm{Lie}(\mathcal{G}_{\mathrm{red}})$.

Then the conclusion of 2.5(2) holds. As a consequence, $\mathcal{G}_{\mathrm{red}}$ is a normal subgroup scheme of $\mathcal{G}$ and $\mathcal{G}/\mathcal{G}_{\mathrm{red}}$ is of multiplicative type.

Let T be a maximal torus of $\mathcal{G}_{\mathrm{red}}$, and let H be the centralizer of T in $\mathcal{G}$. One has $H \cap \mathcal{G}_{\mathrm{red}} = T$. It follows from (b) that any nilpotent element $\mathfrak{n}$ of $\mathrm{Lie}(H)$ is in $\mathrm{Lie}(T)$, hence vanishes. By the following lemma, H is of multiplicative type, and in particular commutative.

Lemma 2.8. *Let H be a connected algebraic group over k . If all the elements of $\mathrm{Lie}(H)$ are semisimple, then H is of multiplicative type.*

Proof. See also [DG, IV, §3, Lemma 3.7]. The Lie algebra $\mathrm{Lie}(H)$ is commutative. Fix $x \in \mathrm{Lie}(H)$, and let us show that it is central in $\mathrm{Lie}(H)$. Since $\mathrm{ad} x$ is semisimple, it suffices to show that, if y is in an eigenspace of $\mathrm{ad} x$, i.e. $[x, y] = \lambda y$, then x and y commute, i.e. $\lambda = 0$. Let W be the vector subspace of $\mathrm{Lie}(H)$ generated by the elements $y^{[p^\ell]}$, $\ell \geq 0$. The elements $y^{[p^\ell]}$ commute. The map $z \mapsto z^{[p]}$ therefore induces a p -linear map from W to itself, injective by assumption. This implies that W has a basis e_i , $1 \leq i \leq N$, consisting of elements such that

$$e_i^{[p]} = e_i, \quad \left(\sum a_i e_i \right)^{[p]} = \sum a_i^p e_i.$$

To see this, we view W as an algebraic group. The morphism $W \rightarrow W$ given by $x \mapsto x^{[p]} - x$ is étale. Let K be its kernel. In any basis it is defined by $d = \dim(W)$ equations of degree p , namely

$$(x^{[p]})_i - x_i = 0, \quad i = 1, 2, \dots, d,$$

with an étale set of solutions and no solution at infinity: the system of homogeneous equations $(x^{[p]})_i = 0$ has no nonzero solutions by assumption. By Bézout, the group of solutions is hence a $(\mathbb{Z}/p)^d$. Let e_i be a basis for it. If the e_i were not linearly independent in W , there would be a linear dependence $\sum_{i \in J} a_i e_i = 0$, with the $a_i \in k$ nonzero, involving a minimal number of e_i . We also have $\sum a_i^p e_i = 0$. Hence, by the minimality of the e_i , for some λ and all i , one has $a_i^p = \lambda a_i$. Rescaling by $\lambda^{1/(p-1)}$, we get $a_i \in \mathbb{F}_p$, which is a contradiction. $\square$

Lemma 2.9. *For $b = (b_i) \in k^N$, define $b^{[p]} := (b_i^p)$. Then any $b \in k^N$ is a linear combination of the $b^{[p^\ell]}$ for $\ell \geq 1$.*

Proof. The $b^{[p^a]}$, $a \geq 0$, are linearly dependent. A linear dependence relation can be written

$$\sum_{j \geq m} c_j b^{[p^j]} = 0 \quad (2.9.1)$$

with $c_m \neq 0$. Extracting p^m -roots, we get

$$\sum_{j \geq 0} d_j b^{[p^j]} = 0, \quad (2.9.2)$$

where $d_j = c_{m+j}^{1/p^m}$. In particular, $d_0 \neq 0$, proving the lemma. $\square$

End of proof of commutativity. From the lemma above, y is a linear combination of the $y^{[p^\ell]}$, $\ell > 0$. The bracket $[y^{[p^\ell]}, x]$ vanishes for $\ell > 0$. Indeed, it is $(\text{ad } y)^{[p^\ell]}(x)$, which vanishes because $[y, [y, x]] = [y, -\lambda y] = 0$. It follows that $[y, x]$ vanishes too.

The p -Lie algebra $\text{Lie}(H)$ is commutative and hence its p -power operation $\text{Lie}(H) \rightarrow \text{Lie}(H)$ is injective. It therefore has a basis e_i such that $e_i^{[p]} = e_i$. The dual of its restricted enveloping algebra is the affine bigebra of the kernel K of the Frobenius morphism $F : H \rightarrow H^{(p)}$, where $H^{(p)}$ is obtained from H by extension of scalars $\lambda \mapsto \lambda^{[p]}$, $k \mapsto k$. It follows that K is a product of μ_p 's.

The same holds for $H^{(p)}$, which is obtained from H using an automorphism of k . The same holds for each $H^{(p^\ell)}$ as well.

For any n , the kernel K_n of the iterated Frobenius map $F^n : H \rightarrow H^{(p^n)}$ is an iterated extension of subgroups of the kernel of the Frobenius of the $H^{(p^i)}$, $i < n$. It is hence of multiplicative type, being an iterated extension of connected groups of multiplicative type [SGA3, XVII, 7.1.1].

The K_n form an increasing sequence. By the proof of [D, Proposition 1.1], there exists a subgroup of multiplicative type M of H containing all K_n . Since H is connected and since M contains all infinitesimal neighborhoods of the identity element, one has $M = H$.

Whenever a group M of multiplicative type acts on a group K , its action on $\text{Lie}(K)$ defines a weight decomposition

$$\text{Lie}(K) = \bigoplus_{\beta \in X(M)} \text{Lie}(K)_\beta. \quad (2.9.3)$$

If $v \in \text{Lie}(K)_\beta$, then $v^{[p]}$ is in $\text{Lie}(K)_{p\beta}$. Indeed, after any extension of scalars R/k , if $m \in M(R)$,

$$m(v^{[p]}) = (m(v))^{[p]} = (\beta(m)v)^{[p]} = \beta(m)^{[p]}v^{[p]}. \quad (2.9.4)$$

Lemma 2.10. *Let M be a multiplicative group acting on a group K , and let (2.9.3) be the weight-space decomposition for the corresponding action on $\text{Lie}(K)$. If $\beta \in X(M)$ is not torsion, then $\text{Lie}(K)_\beta$ consists of nilpotent elements.*

Proof. Indeed, if β is not torsion, the $p^\ell \beta$ are all distinct and $\text{Lie}(K)_{p^\ell \beta}$ must vanish for $\ell \gg 0$. It follows that the elements of $\text{Lie}(K)_\beta$ are nilpotent. $\square$

Let us apply this to the action of T on $\mathcal{G}$ by inner automorphisms.

Lemma 2.11. *If $\beta \in X(T)$ is nonzero, the weight space $\text{Lie}(\mathcal{G})_\beta$ equals $\text{Lie}(\mathcal{G}_{\text{red}})_\beta$.*

Proof. Indeed, $\text{Lie}(\mathcal{G})_\beta$ consists of nilpotent elements. By our assumption 2.7(b), it is contained in $\text{Lie}(\mathcal{G}_{\text{red}})$. $\square$

Let B be a Borel subgroup of $\mathcal{G}_{\text{red}}$ containing T , and let U be its unipotent radical.

Lemma 2.12. *Under the assumption of 2.7, H normalizes U .*

Proof. Let C in $X(T) \otimes \mathbb{R}$ be the cone generated by the positive roots relative to B , and define

$$C^* := C \setminus \{0\}. \quad (2.12.1)$$

As in 2.11, we let T act on $\mathcal{G}$ by conjugation, so that t acts by $g \mapsto tgt^{-1}$. This action induces actions on $\mathfrak{g} := \text{Lie}(\mathcal{G})$, on the affine algebra A of $\mathcal{G}$, on its augmentation ideal $\mathfrak{m}$, and on the dual $\mathfrak{g}^\vee = \mathfrak{m}/\mathfrak{m}^2$ of $\mathfrak{g}$. Similarly, T acts on U , on its Lie algebra $\mathfrak{u}$, on its affine algebra A_U , and on its augmentation ideal $\mathfrak{m}_U$. For the action on affine algebras, $t \in T$ transforms $f(g)$ into $f(t^{-1}gt)$. From these actions we get $X(T)$ -gradings. By 2.7(b),

$$\mathfrak{u} = \bigoplus_{\beta \in C^*} \mathfrak{g}_\beta. \quad (2.12.2)$$

It follows that $\mathfrak{u}^\vee = \mathfrak{m}_U/\mathfrak{m}_U^2$ is the sum of the $(\mathfrak{g}^\vee)_\beta$, for β in the negative $-C^*$ of C^* . For $n > 0$, the weights by which T acts on $\mathfrak{m}_U^n/\mathfrak{m}_U^{n+1}$ are in $-C^*$. Since U is connected, the intersection of the $\mathfrak{m}_U^n$ is reduced to 0, and the weights by which T acts on $\mathfrak{m}_U$ are also in $-C^*$.

Let I be the ideal of A generated by the graded components $\mathfrak{m}_\beta$ of $\mathfrak{m}$ for $\beta \notin -C^*$, and put $A_1 := A/I$. The image in $\mathfrak{m}_U$ of an $\mathfrak{m}_\beta$ as above is contained in $(\mathfrak{m}_U)_\beta$, hence vanishes. It follows that U is contained in the closed subscheme $U_1 = \text{Spec } A_1$ of $\mathcal{G}$ defined by I . As the graded component $\mathfrak{m}^0$ of $\mathfrak{m}$ is contained in I , the graded component A_1^0 of A_1 , the T -invariants, is reduced to the constants. Since T is connected, it follows that U_1 is connected. Since the image of $(\mathfrak{m}_U)_\beta$ in $\mathfrak{m}/\mathfrak{m}^2$ is $(\mathfrak{m}/\mathfrak{m}^2)_\beta$, the image of I in $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{g}^\vee$ is the orthogonal of $\mathfrak{u} \subset \mathfrak{g}$, and the tangent space at the origin of U_1 is $\mathfrak{u}$.

Claim. The subscheme U_1 of $\mathcal{G}$ is a subgroup scheme; that is, the coproduct $\Delta : A \rightarrow A \otimes A$, $f(g) \mapsto f(gh)$, maps I to $I \otimes A + A \otimes I$. Indeed, Δ respects the gradings. For $k \subset A$ the constants, one has $A = k \oplus \mathfrak{m}$, and Δ maps $\mathfrak{m}$ to $(k \otimes \mathfrak{m}) \oplus (\mathfrak{m} \otimes k) \oplus (\mathfrak{m} \otimes \mathfrak{m})$, and hence $\mathfrak{m}_\beta$ maps to

$$(k \otimes \mathfrak{m}_\beta) \oplus (\mathfrak{m}_\beta \otimes k) \oplus \sum_{\beta = \beta' + \beta''} \mathfrak{m}_{\beta'} \otimes \mathfrak{m}_{\beta''}. \quad (2.12.3)$$

Since $-C^*$ is stable under addition, if $\beta = \beta' + \beta''$ and β is not in $-C^*$, one of β' or β'' is not in $-C^*$, and the corresponding $\mathfrak{m}_{\beta'}$ or $\mathfrak{m}_{\beta''}$ is contained in I . The claim follows.

To summarize, U_1 is connected, and the inclusion $U \subset U_1$ induces an isomorphism $\text{Lie}(U) \xrightarrow{\sim} \text{Lie}(U_1)$. Since U is smooth, this implies that $U = U_1$.

Since H centralizes T , the ideal I is stable under H , meaning that H normalizes U . $\square$

Corollary 2.13. *Under the assumption of 2.7, H normalizes $\mathcal{G}_{\text{red}}$.*

Proof. Let B^- be the Borel subgroup of $\mathcal{G}_{\text{red}}$ containing T and opposite to B , and let U^- be its unipotent radical. As U^- , T , and U are normalized by H , the big cell $U^-TU \subset \mathcal{G}_{\text{red}}$ is stable under the conjugation action of H , and so is its schematic closure $\mathcal{G}_{\text{red}}$. $\square$

In what follows, we identify schemes with the corresponding fppf sheaves. A quotient such as $\mathcal{G}/H$ represents the quotient of the sheaf of groups $\mathcal{G}$ by the subsheaf H : $\mathcal{G}$ is an H -torsor over $\mathcal{G}/H$.

Lemma 2.14. *The morphism of schemes*

$$\mathcal{G}_{\text{red}}/T \longrightarrow \mathcal{G}/H \quad (2.14.1)$$

is an isomorphism.

Proof. As T is the intersection of $\mathcal{G}_{\text{red}}$ and H , the morphism (2.14.1), as a morphism of fppf sheaves, is injective. Testing on $\text{Spec}(k)$ and $\text{Spec}(k[\varepsilon]/(\varepsilon^2))$, one sees that it is bijective on points and injective on tangent spaces at each point. It is hence radicial and unramified. Therefore, on

some open set of $\mathcal{G}_{\text{red}}/T$, this morphism is an immersion. The $\mathcal{G}_{\text{red}}$ -homogeneity then shows that it is a closed embedding. Since $\mathcal{G}_{\text{red}}/T$ is smooth and (2.14.1) is bijective on k -points, to prove that the closed embedding (2.14.1) is an isomorphism, it suffices to prove that at each point it induces an isomorphism of tangent spaces. By homogeneity, it suffices to check this at the origin. The tangent space at the origin of $\mathcal{G}_{\text{red}}/T$ is $\text{Lie}(\mathcal{G}_{\text{red}})/\text{Lie}(T)$.

For the tangent space of $\mathcal{G}/H$ we proceed as follows. Let J be the ideal defining H in $\mathcal{G}$, and let η be the ideal defining the origin in $\mathcal{G}/H$. Since $\mathcal{G}$ is an H -torsor over $\mathcal{G}/H$, the pull-back of η/η^2 to $\mathcal{G}$ is J/J^2 , and η/η^2 is the fiber of J/J^2 at the origin e . By [D, 2.15] applied to T acting on $\mathcal{G}$ by conjugation, we have an isomorphism

$$(J/J^2)_e \xrightarrow{\sim} \bigoplus_{\beta \neq 0} (\mathfrak{m}/\mathfrak{m}^2)_{\beta} = (\text{Lie}(\mathcal{G}_{\text{red}})/\text{Lie}(T))^{\vee}.$$

Taking duals, we see that (2.14.1) is an isomorphism near the origin, hence everywhere. $\square$

Corollary 2.15. *Under the assumptions of 2.7, $\mathcal{G}_{\text{red}}$ is normal in $\mathcal{G}$, and there exists in $\mathcal{G}$ a central connected subgroup scheme of multiplicative type M such that the morphism*

$$M \times \mathcal{G}_{\text{red}} \longrightarrow \mathcal{G}$$

realizes $\mathcal{G}$ as a quotient of $M \times \mathcal{G}_{\text{red}}$.

Proof. By 2.14, the product map $\mathcal{G}_{\text{red}} \times H \rightarrow \mathcal{G}$ is onto as a morphism of sheaves. Since both $\mathcal{G}_{\text{red}}$ and H normalize $\mathcal{G}_{\text{red}} \subset \mathcal{G}$, so does $\mathcal{G}$.

To complete the proof of 2.15, and thereby of 2.7, we follow [D, §2.25]. Let M be the subgroup of H which centralizes $\mathcal{G}_{\text{red}}$, as always in the scheme-theoretic sense. By [DG, Corollary 2.4(a), page 476], M is also multiplicative. Since $\mathcal{G}_{\text{red}}$ is reductive, the group scheme $\text{Aut}_T(\mathcal{G}_{\text{red}})$ of automorphisms preserving T is precisely T^{ad} , the image of T in the adjoint group. Hence the conjugation action of H on $\mathcal{G}_{\text{red}}$ gives the exact sequence

$$1 \longrightarrow M \longrightarrow H \longrightarrow T^{\text{ad}} \longrightarrow 1, \quad (2.15.1)$$

and T surjects onto T^{ad} , implying that M and T generate H . Since M is generated by $M_{\text{red}} \subset \mathcal{G}_{\text{red}}$ and M^0 , and since H and $\mathcal{G}_{\text{red}}$ generate $\mathcal{G}$, we see that M^0 and $\mathcal{G}_{\text{red}}$ generate $\mathcal{G}$. Moreover M^0 is central. Thus

$$M^0 \times \mathcal{G}_{\text{red}} \longrightarrow \mathcal{G} \quad (2.15.2)$$

is an epimorphism. This concludes the proof of 2.7 and in particular of 2.5(2). $\square$

Lemma 2.16. *Suppose that H is a maximal connected subgroup scheme of multiplicative type of an algebraic group $\mathcal{G}$. Let $Z_{\mathcal{G}}^0(H)$ be the identity component of the centralizer of H in $\mathcal{G}$, and define $U := Z_{\mathcal{G}}^0(H)/H$. Then the sequence*

$$0 \longrightarrow \text{Lie}(H) \longrightarrow \text{Lie}(Z_{\mathcal{G}}^0(H)) \longrightarrow \text{Lie}(U) \longrightarrow 0 \quad (2.16.1)$$

associated with the central extension

$$1 \longrightarrow H \longrightarrow Z_{\mathcal{G}}^0(H) \longrightarrow U \longrightarrow 1 \quad (2.16.2)$$

is exact.

Proof. Left exactness of (2.16.1) is clear. The maximality of H implies that U is unipotent; see [D, §2.5, page 590].

Embed H in $\mathbf{G}_m^r$ as a subgroup scheme. The quotient $H'' = \mathbf{G}_m^r/H$ is a torus, being a quotient of one. The central extension (2.16.2), by a push-forward, gives a central extension [SGA3, Exposé XVII, Lemma 6.2.4]

$$1 \longrightarrow \mathbf{G}_m^r \longrightarrow E \longrightarrow U \longrightarrow 1 \quad (2.16.3)$$

and a diagram of groups

$$\begin{array}{ccccccc}
& & 1 & & 1 & & 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & H & \longrightarrow & Z_{\mathcal{G}}^0(H) & \longrightarrow & U \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \parallel \\
1 & \longrightarrow & \mathbf{G}_m^r & \longrightarrow & E & \longrightarrow & U \longrightarrow 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
1 & \longrightarrow & H'' & \xlongequal{\quad} & H'' & \longrightarrow & 1 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 1 & & 1 & & 1
\end{array} \tag{2.16.4}$$

Since H'' is multiplicative, every nilpotent in $\mathrm{Lie}(E)$ maps to 0 in $\mathrm{Lie}(H'')$, and hence comes from a nilpotent in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H))$. Since $\mathbf{G}_m^r$ is smooth, [SGA3, Exposé VII, Proposition 8.2] gives from (2.16.3) an exact sequence

$$0 \longrightarrow \mathrm{Lie}(\mathbf{G}_m^r) \longrightarrow \mathrm{Lie}(E) \longrightarrow \mathrm{Lie}(U) \longrightarrow 0. \tag{2.16.5}$$

Since U is unipotent, any element $z \in \mathrm{Lie}(U)$ is nilpotent. Let $z' \in \mathrm{Lie}(E)$ be a lift of z . The Jordan decomposition makes sense for any p -Lie algebra over a perfect field k and uses only the p -power map; see, for example, [W2, Corollary 4.5.9, page 135]. Thus, by using the Jordan decomposition of the lift z' in $\mathrm{Lie}(E)$, and noting that the semisimple part gets mapped to zero in $\mathrm{Lie}(U)$, we can assume that z' can also be chosen to be nilpotent.

Since every nilpotent in $\mathrm{Lie}(E)$ comes from a nilpotent in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H))$, we conclude that z is lifted to a nilpotent in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H))$. This implies that (2.16.1) is also right exact. $\square$

Lemma 2.17. *Suppose that $\mathcal{G}$ is a subgroup scheme of G which is infinitesimally saturated in G . Then the subgroup scheme $Z_{\mathcal{G}}^0(H)$ is infinitesimally saturated in G . In particular, every nonzero nilpotent in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H))$ lies in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H)_{\mathrm{red}})$.*

Proof. If $\mathfrak{n} \in \mathrm{Lie}(Z_{\mathcal{G}}^0(H))$ is nilpotent, the map $\rho : t \mapsto \exp(t\mathfrak{n}) : \mathbf{G}_a \rightarrow G$ factors through $\mathcal{G}$. Since H is central in $Z_{\mathcal{G}}^0(H)$, the action of H by inner automorphisms fixes $\mathfrak{n}$. Since the map $\exp$ is compatible with conjugation, the entire curve ρ is fixed by H . Therefore ρ factors through $Z_{\mathcal{G}}(H)$, and hence through $Z_{\mathcal{G}}^0(H)_{\mathrm{red}}$, since $\mathbf{G}_a$ is reduced and connected. Thus $\mathfrak{n} \in \mathrm{Lie}(Z_{\mathcal{G}}^0(H)_{\mathrm{red}})$. $\square$

We have the following extension of [D, Lemma 2.7, Corollary 2.11].

Lemma 2.18. *Let $\mathcal{G}$ be as in 2.17. Let $H \subset \mathcal{G}$ be a maximal connected subgroup scheme of multiplicative type. Then the central extension (2.16.2) splits and U is a smooth unipotent group.*

Proof. We claim that U is smooth. By Lemma 2.16, every element $z \in \mathrm{Lie}(U)$ comes from a nilpotent $\mathfrak{n} \in \mathrm{Lie}(Z_{\mathcal{G}}^0(H))$. By 2.17, $\mathfrak{n}$ is in $\mathrm{Lie}(Z_{\mathcal{G}}^0(H)_{\mathrm{red}})$. Its image z is hence in $\mathrm{Lie}(U_{\mathrm{red}})$. Thus $\mathrm{Lie}(U) = \mathrm{Lie}(U_{\mathrm{red}})$, proving the claim.

Now we have a central extension (2.16.2) with the added feature that U is smooth. By [SGA3, Exposé XVII, Theorem 6.1.1], it follows that (2.16.2) splits uniquely and

$$Z_{\mathcal{G}}^0(H) = H \times U. \tag{2.18.1}$$

$\square$

Remark 2.19. Recall that in 2.2, for the existence of an exponential map (2.2.2), we assumed that the covering morphism $\tilde{G} \rightarrow G'$ is étale, which therefore became a part of the standing assumption (2.3). The case which gets excluded is when the simply connected cover of the derived group has factors of $\mathrm{SL}(p)$. For instance, when G' is simple with $p = h_G$, the only case excluded is $G' = \mathrm{PGL}(p)$.

Let G be a reductive group for which $p = h_G$. The tables of Coxeter numbers of simple groups show that except for type A , where $h_{\mathrm{SL}(n)} = n$, Coxeter numbers are even greater than 2. It follows that $\tilde{G}$ is a product of $\mathrm{SL}(p)$'s and other simple factors with p larger than their Coxeter numbers. Even when the morphism $\tilde{G} \rightarrow G'$ is not étale, we can still define the notions of saturation, respectively infinitesimal saturation, of subgroup schemes $\mathcal{G} \subset G$ as follows.

Say that $\mathcal{G} \subset G$ is saturated, respectively infinitesimally saturated, if the inverse image of $\mathcal{G}$ in $\tilde{G}$ is saturated, respectively infinitesimally saturated. With this definition, Theorem 2.5 remains true for $p \geq h_G$.

This notion of saturation, respectively infinitesimal saturation, can also be seen in terms of suitably defined t -power maps and exponential maps. We restrict ourselves to the case when $G' = \mathrm{PGL}(p)$. At the level of Lie algebras, the induced morphism

$$\mathfrak{sl}(p) \longrightarrow \mathfrak{pgl}(p) \quad (2.19.1)$$

is a radicial map on the locus of nilpotent elements. If $A \in \mathfrak{gl}(p)$ is a matrix representing an element of $\mathfrak{pgl}(p)$, it is nilpotent if all but the constant coefficient of the characteristic polynomial vanish, i.e. $\mathrm{Tr}(\wedge^i A) = 0$ for $1 \leq i < p$, and the characteristic polynomial reduces to $T^p - \det(A)$. This condition is stable under $A \mapsto A + \lambda I$, as

$$(T - \lambda)^p - \det(A) = T^p - (\det(A) + \lambda^p) = T^p - \det(A + \lambda I).$$

We get the unique lift $\tilde{A} \in \mathfrak{sl}(p)$ by taking $\tilde{A} := A - \det(A)^{1/p} I$. Now one has $\mathrm{Tr}(\wedge^i \tilde{A}) = 0$ for $1 \leq i \leq p$, and hence we can define the exponential morphism $\mathbf{G}_a \rightarrow \mathrm{PGL}(p)$ as

$$t \longmapsto \exp(tA) := \exp(t\tilde{A}). \quad (2.19.2)$$

Likewise, in the case of unipotents in the group $\mathrm{PGL}(p)$, the restriction of this map to the locus of unipotent elements

$$\mathrm{SL}(p)_u \longrightarrow \mathrm{PGL}(p)_u \quad (2.19.3)$$

is radicial. Thus any unipotent $u \in \mathrm{PGL}(p)$ has a unique unipotent lift $\tilde{u} \in \mathrm{SL}(p)$, and one can define the t -power map $\mathbf{G}_a \rightarrow \mathrm{PGL}(p)$ as

$$t \longmapsto u^t := \mathrm{image}(\tilde{u}^t). \quad (2.19.4)$$

With the notions of t -power map and exponential morphisms in place, we can define the notions of saturation, respectively infinitesimal saturation, of subgroup schemes $\mathcal{G} \subset G'$ exactly as in Definition 2.4 using (2.19.4), respectively (2.19.2), and these coincide with the definitions made above.

We note however that these punctual maps, i.e. maps defined for each nilpotent A , respectively each unipotent u , are not induced by a morphism from $\mathfrak{pgl}(p)_{\mathrm{nilp}}$ to $\mathrm{PGL}(p)_u$, respectively $\mathbb{A}^1 \times \mathrm{PGL}(p)_u \rightarrow \mathrm{PGL}(p)_u$, $(t, u) \mapsto u^t$, is not a morphism.

3 Completion of proof of the structure theorem 2.5

We begin by stating a general result on root systems whose proof is given in the appendix.

Proposition 3.1. *Let R be an irreducible root system with Coxeter number h , and let X be the lattice spanned by R . Let $\varphi : X \rightarrow \mathbb{R}/\mathbb{Z}$ be a homomorphism. Then there exists a basis B for R such that, if $\alpha \in R$ satisfies $\varphi(\alpha) \in (0, 1/h) \bmod \mathbb{Z}$, then α is positive with respect to B .*

3.2. Assumption 2.3 on the reductive group G continues to be in force. Let $\mathcal{G}$ be a subgroup scheme of G , and let H be a maximal connected subgroup scheme of multiplicative type of $\mathcal{G}$. For the existence of H , see [D, Proposition 2.1].

Let $X(H)$ be the group of characters of H . The action of H on $\mathrm{Lie}(G)$ by conjugation gives an $X(H)$ -gradation

$$\mathrm{Lie}(G) = \mathrm{Lie}(Z_G^0(H)) \oplus \bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)_\alpha.$$

Recall that if M is a connected group scheme of multiplicative type over k , the group of characters $X(M)$ is finitely generated and does not have torsion prime to p . As k is algebraically closed, this means that M is a product of subgroups isomorphic to $\mathbf{G}_m$ or μ_{p^j} with $j > 0$.

Corollary 3.3. *Cf. [D, Lemma 2.12]. Let M be a connected subgroup scheme of multiplicative type of G , and suppose that $p > h_G$. The action of M on $\mathrm{Lie}(G)$ by conjugation gives an $X(M)$ -gradation*

$$\mathrm{Lie}(G) = \bigoplus_{\alpha \in X(M)} \mathrm{Lie}(G)_\alpha.$$

If $\alpha \neq 0$, then for $\gamma \in \mathrm{Lie}(G)_\alpha$ one has $\gamma^{[p]} = 0$. In particular, if $M = H$ and $\gamma \in \mathrm{Lie}(G)_\alpha$, we have $\gamma^{[p]} = 0$.

Proof. If α is of infinite order, it suffices to apply 2.10. If α is of finite order, there is a subgroup N of M isomorphic to μ_{p^j} to which α restricts nontrivially, and it suffices to prove 3.3 for N and the restriction of α to N . Choose an isomorphism of $X(N)$ with $X(\mu_{p^j})$ such that the image of α in $\mathbb{Z}/p^j$ is of the form p^b , with $0 \leq b < j$.

In a smooth algebraic group, the maximal connected subgroups of multiplicative type are maximal tori. Indeed, if T is such a maximal subgroup, its centralizer $Z_G(T)$ is smooth, being the fixed locus of a linearly reductive group acting on a smooth variety; see for example [DG, Theorem 2.8, Chapter II, §5]. Define $U := Z_G(T)^0/T$. The group scheme U is smooth, as a quotient of $Z_G(T)^0$, which is smooth, and is unipotent by maximality of T . By [SGA3, Exposé XVII, Theorem 6.1.1], we have a splitting $Z_G(T)^0 \simeq T \times U$, and hence T is smooth.

In our case, N is hence contained in a maximal torus T of G . Let

$$\varphi : X(T) \longrightarrow \mathbb{R}/\mathbb{Z} \tag{3.3.1}$$

be the composite of $X(T) \rightarrow X(N) = \mathbb{Z}/p^j$ and of the inclusion $x \mapsto x/p^j$ of $\mathbb{Z}/p^j$ in $\mathbb{R}/\mathbb{Z}$. The N -weight space $\mathrm{Lie}(G)_\alpha$ is the sum of T -weight spaces $\mathrm{Lie}(G)_\beta$, for β a root such that $\varphi(\beta) = 1/p^{j-b}$. As

$$\frac{1}{p^{j-b}} \leq \frac{1}{p} < \frac{1}{h},$$

we may apply 3.1, to conclude that $\mathrm{Lie}(G)_\alpha$ is contained in the Lie algebra of the unipotent radical of a Borel subgroup of G . In particular, it consists of nilpotent elements, proving 3.3. $\square$

3.4. The proof of 2.5 now follows [D, pages 594–599] verbatim with a single alteration. Recall that in [D], the group G was the linear group $\mathrm{GL}(V)$ and the condition on the characteristic was $p > \dim(V)$. For an arbitrary connected reductive G , this condition is now replaced by $p > h_G$, which makes 3.3 applicable.

Remark 3.5. If $G = \mathrm{GL}(V)$, one has $h_G = \dim(V)$. In the case $G = \prod \mathrm{GL}(V_i)$, with $p > \dim(V_i)$ for each i , the case $p > h_G$ of Theorem 2.5 gives us [D, Theorem 1.7].

Example 3.6 (Brian Conrad). Here is an example, in any characteristic $p > 0$, of a connected group of multiplicative type M acting on a reductive group G , and of a nontrivial character α of M , such that the weight space $\mathrm{Lie}(G)_\alpha$ contains elements which are not nilpotent. We take $G = \mathrm{SL}_p$, so that $h_G = p$, and $M = \mu_{p^2}$, and for α the character $\zeta \mapsto \zeta^p$.

We embed M in the maximal torus of diagonal matrices of SL_p by

$$\zeta \mapsto \mathrm{diag}(\zeta^0, \zeta^{-p}, \zeta^{-2p}, \dots, \zeta^{-(p-1)p}).$$

The restriction to M of each simple root and of the lowest root is the character $\alpha : \zeta \mapsto \zeta^p$, and $\mathrm{Lie}(G)_\alpha$ is the sum of the corresponding root spaces. In the standard visualization of SL_p , this weight space inside $\mathrm{Lie}(G) = \mathfrak{sl}_p$ is the span of the superdiagonal entries and the lower-left entry.

A sum of nonzero elements in those root lines contributing to $\mathrm{Lie}(G)_\alpha$ is a $p \times p$ -matrix $X \in \mathfrak{sl}_p$ which satisfies

$$X(e_1) = t_p e_p, \quad X(e_2) = t_1 e_1, \quad \dots, \quad X(e_p) = t_{p-1} e_{p-1}. \quad (3.6.1)$$

Iterating p times gives $X^p = \mathrm{diag}(t, \dots, t)$, with $t := \prod t_j \neq 0$. Hence $X \in \mathrm{Lie}(G)_\alpha$ is not nilpotent.

Example 3.7. A variant of the above example leads to an example of an infinitesimally saturated group scheme $\mathcal{G} \subset \mathrm{SL}(V)$, with $\dim(V) = p$, such that V is an irreducible representation and $\mathcal{G}_{\mathrm{red}}$ is a unipotent group. This in particular implies that $\mathcal{G}_{\mathrm{red}}$ is not normal in $\mathcal{G}$.

Let V be the affine algebra of μ_p , that is

$$V = \mathcal{O}(\mu_p) := \frac{k[u]}{(u^p - 1)}. \quad (3.7.1)$$

The vector space V admits the basis $\{u^i \mid i \in \mathbb{Z}/p\}$. The multiplicative group $\mathcal{O}(\mu_p)^*$ acts by multiplication on V . For $f \in \mathcal{O}(\mu_p)^*$, f^p is constant. Define N_f to be the constant value of f^p . It is in fact the norm of f . The action of $\mathcal{O}(\mu_p)^*$ on V induces an action of the subgroup $N \subset \mathcal{O}(\mu_p)^*$ for which $N_f = 1$.

On V we also have the action of μ_p by translations, and this action normalizes the group $\mathcal{O}(\mu_p)^*$ and its subgroup N . Consider the group scheme

$$\mathcal{G} := \mu_p \ltimes N. \quad (3.7.2)$$

We make a few observations on $\mathcal{G}$:

- (1) The group $\mathcal{G}$ is infinitesimally saturated.
- (2) In $\mathcal{G}$, consider the first factor μ_p and $\{u^i \mid i \in \mathbb{Z}/p\} \simeq \mathbb{Z}/p$. These subgroups generate a subgroup H , which is a central extension of $\mu_p \ltimes \mathbb{Z}/p$ by μ_p .
- (3) The representation V is irreducible as an H -module. Indeed, a μ_p -submodule is generated by some of the u^i , and if nonzero and $\mathbb{Z}/p$ -stable, contains all of them. A fortiori, V is an irreducible $\mathcal{G}$ -module.
- (4) The reduced group $\mathcal{G}_{\mathrm{red}}$ can be identified with the unipotent group $\{f \in \mathcal{O}(\mu_p)^* \mid f(1) = 1\}$.
- (5) The subgroup $\mathcal{G}_{\mathrm{red}}$ is not normal, as the point 1 of μ_p is not invariant under translations.

4 Semi-simplicity statements

Let G be a reductive group. Let C be an algebraic group and $\rho : C \rightarrow G$ a morphism.

Definition 4.1 ([S2, page 20]). *One says that ρ is cr if, whenever ρ factors through a parabolic P of G , it factors through a Levi subgroup of P .*

When $G = \mathrm{GL}(V)$, ρ is cr if and only if the representation V of C is completely reducible, or equivalently semisimple, and hence the terminology.

The property of ρ being cr depends only on the subgroup scheme of G which is the schematic image of C . It in fact only depends on the image of C in the adjoint group G^{ad} . Indeed, the parabolic subgroups of G are the inverse images of the parabolic subgroups of G^{ad} , and similarly for the Levi subgroups. A subgroup scheme $\mathcal{G}$ of G will be called cr if its inclusion in G is so.

For an irreducible root system R , let α_0 be the highest root and let $\sum n_i \alpha_i$ be its expression as a linear combination of the simple roots. The characteristic p of k is called good for R if p is larger than each n_i . For a general root system R , p is good if it is so for each irreducible component of R .

Proposition 4.2. *Suppose C is an extension*

$$1 \longrightarrow B \longrightarrow C \longrightarrow A \longrightarrow 1 \quad (4.2.1)$$

with A^0 of multiplicative type and A/A^0 a finite group of order prime to p , and suppose that p is good. Let $\rho : C \rightarrow G$ be a morphism. If the restriction of ρ to B is cr, then ρ is cr.

We do not know whether the proposition holds without the assumption that p is good.

Proof. Let P be a parabolic subgroup, U its unipotent radical, and $\mathfrak{u}$ the Lie algebra of U . The parabolic P is said to be *restricted* if the nilpotence class of U is less than p . If P is a maximal parabolic corresponding to a simple root α , the nilpotence class of U is the coefficient of α in the highest root. It follows that p is good if and only if all the maximal parabolic subgroups are restricted. By [Sei, Proposition 5.3], credited by the author to Serre, see 6.6, if P is restricted, one obtains by specialization from characteristic zero a P -equivariant isomorphism

$$\exp : (\mathfrak{u}, \circ) \xrightarrow{\sim} U \quad (4.2.2)$$

from $\mathfrak{u}$, endowed with the Campbell–Hausdorff group law, to U .

We will first show that whenever ρ factors through a restricted parabolic subgroup P as above, if the restriction of ρ to B factors through some Levi subgroup of P , then ρ itself factors through some Levi subgroup of P .

The group $U(k)$ acts on the right on the set $\mathcal{L}(k)$ of Levi subgroups of P by

$$u \in U(k) \quad \text{acts by} \quad L \longmapsto u^{-1}Lu.$$

This action turns $\mathcal{L}(k)$ into a $U(k)$ -torsor. This expresses the fact that two Levi subgroups are conjugate by a unique element of $U(k)$. The group $P(k)$ acts on $\mathcal{L}(k)$ and on $U(k)$ by conjugation. This turns $\mathcal{L}(k)$ into an equivariant $U(k)$ -torsor.

We will need a scheme-theoretic version of the above. Fix a Levi subgroup L_0 . Let $\mathcal{L}$ be the trivial U -torsor, that is, U with the right action of U by right translations. We have the family of Levi subgroups

$$L_u := u^{-1}L_0u$$

parametrized by $\mathcal{L} = U$. We let P act on U by conjugation, and on $\mathcal{L}$ as follows: if $p = v\ell$ in $P = UL_0$, then p acts on $\mathcal{L} = U$ by

$$u \longmapsto v^{-1}pup^{-1}.$$

This turns $\mathcal{L}$ into an equivariant U -torsor. When we pass to k -points, and attach to $u \in \mathcal{L}$ the Levi subgroup $u^{-1}L_0u$, we recover the previously described situation.

The morphism $\rho : C \rightarrow P$ turns $\mathcal{L}$ into an equivariant U -torsor. A point x of $\mathcal{L}$, corresponding to a Levi subgroup L_x , is fixed by C , scheme-theoretically, if and only if ρ factors through L_x . This expresses the fact that a Levi subgroup is its own normalizer in P .

We want to prove that if B has a fixed point in $\mathcal{L}$, then so does C . Let U^B be the subgroup of U fixed by B , for the conjugation action. If B has a fixed point in $\mathcal{L}$, the fixed locus $\mathcal{L}^B$ is a U^B -torsor. As B is a normal subgroup of C , the group C acts on $\mathcal{L}^B$ and on U^B , and the action factors through A .

The isomorphism $\exp : (\mathfrak{u}, \circ) \rightarrow U$ is compatible with the action of B by conjugation. Hence it induces an isomorphism

$$(\mathfrak{u}^B, \circ) \xrightarrow{\sim} U^B.$$

Let $Z^i(\mathfrak{u}^B)$ be the central series of $\mathfrak{u}^B$, and define

$$Z^i(U^B) := \exp(Z^i(\mathfrak{u}^B)).$$

The isomorphism $\exp$ induces an isomorphism between the vector group

$$\mathrm{Gr}_Z^i(\mathfrak{u}^B)$$

and

$$Z^i(U^B)/Z^{i+1}(U^B),$$

compatible with the action of A . On $\mathrm{Gr}_Z^i(\mathfrak{u}^B)$, this action is linear.

The assumption on A amounts to saying that A is linearly reductive, that is, all its representations are semi-simple. Equivalently, if k is the trivial representation, any extension

$$0 \longrightarrow V \xrightarrow{a} E \xrightarrow{b} k \longrightarrow 0 \quad (4.2.3)$$

splits. Passing from E to $b^{-1}(1)$, such extensions correspond to A -equivariant V -torsors, and the extension splits if and only if A has a scheme-theoretic fixed point on the corresponding torsor.

Define U_i^B to be $U^B/Z^i(U^B)$, and $\mathcal{L}_i^B$ to be the U_i^B -torsor obtained from $\mathcal{L}^B$ by pushing out by $U^B \rightarrow U_i^B$. We prove by induction on i that A has a fixed point on $\mathcal{L}_i^B$.

As U_1^B is trivial, the case $i = 1$ is trivial. If x is a fixed point of A in $\mathcal{L}_i^B$, the inverse image of x in $\mathcal{L}_{i+1}^B$ is an equivariant A -torsor on

$$\mathrm{Gr}_Z^i(\mathfrak{u}^B) \simeq Z^i(U^B)/Z^{i+1}(U^B).$$

By linear reductivity, A has a fixed point on the inverse image. As the central descending series of $\mathfrak{u}$, and hence of $\mathfrak{u}^B$, terminates, this proves 4.2 for restricted parabolic subgroups.

We now prove 4.2 by induction on the dimension of G . Suppose that ρ factors through a proper parabolic subgroup P . As p is good, there exists a restricted proper parabolic Q containing P , and P is the inverse image, by the projection $Q \rightarrow Q/R_u(Q)$, of a parabolic P' of $Q/R_u(Q)$. Let L be a Levi subgroup of Q through which ρ factors, and let P'_L be the parabolic subgroup of L obtained as the inverse image of P' by the isomorphism

$$L \xrightarrow{\sim} Q/R_u(Q).$$

Levi subgroups of P'_L are Levi subgroups of P , and it remains to apply the induction hypothesis to L , for which p is good too. $\square$

Remark 4.3. For several results related to 4.2 but in the setting of reduced subgroups, see [BMR1, Theorem 3.10] and [BMR2, Theorem 1.1 and Corollary 3.7].

Fix, in the reductive group G , a maximal torus T , and a system of simple roots corresponding to a Borel subgroup B containing T . Let U be the unipotent radical of B .

Definition 4.4 (cf. [Dy], [S1], [IMP]). *The Dynkin height $\text{ht}_G(V)$ of a representation V of G is the largest among*

$$\left\{ \sum_{\alpha > 0} \langle \lambda, \alpha^\vee \rangle \right\},$$

where λ runs over the weights for the action of T on V .

This notion and this terminology go back to Dynkin [Dy, pages 331–332], where it is called “height”, as in [IMP], while in Serre ([S1] and [S2]), it is simply $n(V)$. If V is an irreducible representation, with dominant weight λ^+ and smallest weight λ^- , it is the sum of the coefficients of $\lambda^+ - \lambda^-$, expressed as a linear combination of the simple roots.

It follows that the product in $\text{End}(V)$ of the action of $\text{ht}_G(V) + 1$ elements of $\text{Lie}(U)$ vanishes and, for $\mathfrak{n}$ nilpotent in $\text{Lie}(G)$, one has

$$\mathfrak{n}^{\text{ht}_G(V)+1} = 0 \quad \text{in } \text{End}(V).$$

4.5. The representation $\rho : G \rightarrow \text{GL}(V)$ is said to be of *low height* if $p > \text{ht}_G(V)$. By [S2, Theorem 6, page 25], representations of low height are semi-simple. One can show that if G admits a representation V of low height which is almost faithful, meaning that its kernel is of multiplicative type, then G satisfies Assumption 2.3. That $p \geq h_G$ results from the more precise statement that $\text{ht}_G(V) \geq h_G - 1$, [S3, (5.2.4), page 213]. For the property that $\tilde{G}/G^0$ is étale, one uses the fact that the non-trivial irreducible representation of $\text{PGL}(p)$ of the smallest height is the adjoint representation, which is of height $2p - 2$.

We now assume that V is of low height, and that Assumption 2.3 holds for G . It follows that any nilpotent $\mathfrak{n}$ in $\text{Lie}(G)$ satisfies $\mathfrak{n}^{[p]} = 0$, and further, the exponential map (2.2.2) is defined. The image $d\rho(\mathfrak{n})$ of $\mathfrak{n}$ in $\text{Lie}(\text{GL}(V)) = \text{End}(V)$ also has a vanishing p -th power; hence $\exp(d\rho(\mathfrak{n})t)$ is defined. By [S2, Theorem 5, page 24], one has the following compatibility statement.

4.6. Compatibility. If $\mathfrak{n}$ in $\text{Lie}(G)$ is nilpotent, then

$$\rho(\exp(t\mathfrak{n})) = \exp(t d\rho(\mathfrak{n})). \quad (4.6.1)$$

As a consequence, if $u^{[p]} = 1$ in G , then

$$\rho(u^t) = \rho(u)^t. \quad (4.6.2)$$

The following theorem is a schematic analogue of [BT], for p large enough.

Theorem 4.7. *Suppose that the reductive group G admits a low-height almost faithful representation $\rho : G \rightarrow \text{GL}(V)$, and that $p > h_G$. Then, for any non-trivial unipotent subgroup U of G , there exists a proper parabolic subgroup P of G , containing the normalizer $N_G(U)$ of U , and whose unipotent radical contains U .*

The condition $p > h_G$ implies that G satisfies Assumption 2.3. If G is simple simply connected, it implies the existence of an almost faithful low-height representation except for G of type F_4 , E_6 , E_7 , or E_8 , in which case the lowest height of a non-trivial representation and the Coxeter number are respectively $16 > 12$, $16 > 12$, $27 > 18$, and $58 > 30$. For these groups, we do not know whether the conclusion of the theorem is valid assuming only $p > h_G$.

Proof. Let V^U be the invariants of U acting on V . It is not zero, because U is unipotent. It is not V , because the representation V is almost faithful, hence faithful on U . It does not have a U -stable supplement V' in V , because U would have invariants in V' .

Let H be the subgroup scheme of G which stabilizes V^U . It contains the normalizer $N_G(U)$ of U . It is a doubly saturated subgroup scheme of G . Indeed, if $h \in H(k)$ is of order p , by (4.6.2),

$$\rho(h^t) = \rho(h)^t = \sum_{i < p} \binom{t}{i} (\rho(h) - 1)^i,$$

which stabilizes V^U ; and similarly, if $\mathfrak{n} \in \text{Lie}(H)$ is nilpotent, then the $\exp(t\mathfrak{n})$ are in H .

As $p > h_G$, Theorem 2.5 ensures that H_{red}^0 is a normal subgroup scheme of H , and that the quotient H/H_{red}^0 is an extension of a finite group of order prime to p by a group of multiplicative type. It follows that $U \subset H_{\text{red}}^0$.

Lemma 4.8. *The group H_{red}^0 is not reductive.*

Proof. If it were, V would be a representation of low height of H_{red}^0 , [S2, Corollary 1, page 25], hence a semi-simple representation of H_{red}^0 , and V^U would have in V an H_{red}^0 -stable supplement. As V does not admit a supplement stable under $U \subset H_{\text{red}}^0$, this is absurd. $\square$

If S is a doubly saturated subgroup scheme of G , we will call $R_u(S_{\text{red}}^0)$ the unipotent radical of S and denote it simply by $R_u(S)$. By 2.5, it is a normal subgroup of S , and $S/R_u(S_{\text{red}}^0)$ does not contain any normal unipotent subgroup. This justifies the terminology.

Define $U_1 := R_u(H)$. By 4.8, it is a non-trivial unipotent subgroup of G , and we can iterate the construction. We define, for $i \geq 1$,

$$H_i := \text{stabilizer of } V^{U_i} \subset V, \quad U_{i+1} := R_u(H_i).$$

One has $U \subset H_{\text{red}}^0$ and

$$U \subset N_G(U) \subset H \subset N_G(U_1) \subset H_1 \subset N_G(U_2) \subset H_2 \subset \cdots. \quad (4.8.1)$$

The $(H_i)_{\text{red}}^0$ form an increasing sequence of smooth connected subgroups of G . It stabilizes, hence so do the sequences of the U_i and of the H_i . If $H_i = H_{i+1}$, one has $H_i = N_G(U_{i+1}) = H_{i+1}$ and

$$U_{i+1} = R_u(H_i) = R_u(H_{i+1}) = R_u(N_G(U_{i+1})_{\text{red}}^0).$$

By [BT, Proposition 2.3, page 99], or for example [H, Section 30.3, Proposition on page 186], this implies that $N_G(U_{i+1})_{\text{red}} = (H_{i+1})_{\text{red}}$ is a proper parabolic subgroup of G . Call it Q . A parabolic subgroup of G is its own normalizer scheme, cf. [SGA3, XII, 7.9], [CGP, page 469]. As $(H_{i+1})_{\text{red}} = Q$ is normal in H_{i+1} , it follows that $H_{i+1} = Q$ and that $N_G(U) \subset Q$.

A Levi subgroup L of Q is a reductive subgroup of G , hence satisfies the assumptions of 4.7. It is isomorphic to $Q/R_u(Q)$. If U is not contained in $R_u(Q)$, we can repeat the argument for the image $\bar{U}$ of U in $Q/R_u(Q)$, which is isomorphic to L . One obtains a proper parabolic subgroup of $Q/R_u(Q)$ which contains the normalizer of $\bar{U}$. Its inverse image in Q is a parabolic subgroup, properly contained in Q and containing the normalizer of U . Iterating, one eventually finds a parabolic P containing $N_G(U)$ and such that $U \subset R_u(P)$. $\square$

Remark 4.9. Let $\tilde{G}$ be the simply connected central extension of the derived group G' , and let $\tilde{G} = \prod_i \tilde{G}_i$ be its simple factors. A representation of low height V of G is almost faithful if and only if its restriction to each $\tilde{G}_i$ is not trivial. It suffices to check this for each $\tilde{G}_i$ separately. Thus we may assume that G is simply connected. The existence of a non-trivial V of low height implies that $p > 2$ for G of type B_n, C_n with $n \geq 2$, or F_4 , and $p > 3$ for G of type G_2 . Let $\bar{G}$ be the image of G in $\text{GL}(V)$. If V is non-trivial, $u : G \rightarrow \bar{G}$ is an isogeny. We want to show that it is a central isogeny. If it is not, the structure of isogenies, [SGA3, XXII, 4.2.13], shows that $\ker(u)$ contains the kernel of the Frobenius. The weights of V are the p -th powers and $\text{ht}_G(V)$ is a multiple of p , contradicting the low-height assumption.

4.10. If G is a reductive group for which Assumption 2.3 holds, and $\mathcal{G}$ is a subgroup scheme of G , the *double saturation* $\mathcal{G}^*$ of $\mathcal{G}$ is the smallest doubly saturated subgroup scheme of G containing $\mathcal{G}$. It is the intersection of the doubly saturated subgroup schemes containing $\mathcal{G}$, and is obtained from $\mathcal{G}$ by iterating the construction of taking the group generated by $\mathcal{G}$, the additive groups $\exp(\mathfrak{t}\mathfrak{n})$ for $\mathfrak{n}$ nilpotent in $\mathrm{Lie}(\mathcal{G})$, and u^t for u of order p in $\mathcal{G}(k)$.

Corollary 4.11. *Let V be a low-height almost faithful representation of a reductive group G . Assume that $p > h_G$. Let $\mathcal{G}$ be a subgroup scheme of G , and let $\mathcal{G}^*$ be its double saturation. Then the following conditions are equivalent:*

- (i) V is a semi-simple representation of $\mathcal{G}$;
- (ii) V is a semi-simple representation of $\mathcal{G}^*$;
- (iii) $\mathcal{G}$ is cr in G ;
- (iv) $\mathcal{G}^*$ is cr in G ;
- (v) the unipotent radical of $(\mathcal{G}^*)_{\mathrm{red}}^0$ is trivial.

Proof. (i) $\Leftrightarrow$ (ii): if W is a subspace of V , the stabilizer in G of W is doubly saturated, as we saw in the beginning of the proof of 4.7. If $\mathcal{G}$ stabilizes W , it follows that $\mathcal{G}^*$ also stabilizes W : the lattice of sub-representations of V is the same for $\mathcal{G}$ and $\mathcal{G}^*$, hence the claim.

(iii) $\Leftrightarrow$ (iv): similarly, the parabolic subgroups of G and their Levi subgroups are doubly saturated, hence contain $\mathcal{G}$ if and only if they contain $\mathcal{G}^*$.

Not (v) $\Rightarrow$ not (ii): let U be the unipotent radical of $(\mathcal{G}^*)_{\mathrm{red}}^0$. If it is non-trivial, $V^U \neq V$, because V is faithful on U and does not have a U -stable supplement. As U is normal in $\mathcal{G}^*$, by 2.5, V^U is a sub-representation for the action of $\mathcal{G}^*$ on V . This contradicts the semi-simplicity of V .

(v) $\Rightarrow$ (ii): the representation V of the reductive group $(\mathcal{G}^*)_{\mathrm{red}}^0$ is of low height, hence semi-simple. By 2.5, $(\mathcal{G}^*)_{\mathrm{red}}^0$ is a normal subgroup of $\mathcal{G}^*$, and the quotient A is linearly reductive. If W is a sub- $\mathcal{G}^*$ -representation of V , A acts on the affine space of $(\mathcal{G}^*)_{\mathrm{red}}^0$ -invariant retractions $V \rightarrow W$. It has a fixed point, whose kernel is a supplement to W .

Not (v) $\Rightarrow$ not (iv): let U be the unipotent radical of $(\mathcal{G}^*)_{\mathrm{red}}^0$. If it is not trivial, there exists a parabolic P containing its normalizer, hence $\mathcal{G}^*$, and the unipotent radical of P contains U , by 4.7. Thus no Levi subgroup of P can contain $\mathcal{G}^*$.

(v) $\Rightarrow$ (iv): by [S2, Theorem 7, page 26], $(\mathcal{G}^*)_{\mathrm{red}}^0$ is cr in G , and one applies 4.2. $\square$

Corollary 4.12. *Let $\rho : G \rightarrow \mathrm{GL}(V)$ be an almost faithful low-height representation, and let $v \in V$ be an element such that the G -orbit of v in V is closed. Then there exists a connected central subgroup scheme of multiplicative type $M \subset G_v^0$ and a surjective homomorphism*

$$M \times G_{v,\mathrm{red}}^0 \longrightarrow G_v^0.$$

Proof. The orbit being closed in V , and hence affine, the reduced stabilizer $G_{v,\mathrm{red}}$ is reductive, [Bo]. Since $p \geq h_G$, and since stabilizers are doubly saturated and 2.5(2) holds for $\mathcal{G} = G_v$, we get the required result. $\square$

We now observe that the results of [D, Section 6] can be obtained as a consequence of 4.11. Note that by the remarks in [D, page 607] it suffices to prove the semisimplicity results in the case when k is algebraically closed.

Theorem 4.13. *Let $\mathcal{G}$ be an algebraic group. Let $(V_i)_{i \in I}$ be a finite family of semi-simple $\mathcal{G}$ -modules and let m_i be integers ≥ 0 . If*

$$\sum_i m_i (\dim V_i - m_i) < p, \tag{4.13.1}$$

then the $\mathcal{G}$ -module

$$\bigotimes_j \bigwedge^{m_j} V_j$$

is semi-simple.

Proof. Let $G = \prod_j \mathrm{GL}(V_j)$ and $V = \bigotimes_j \bigwedge^{m_j} V_j$. Then (4.13.1) is simply the inequality $p > \mathrm{ht}_G(V)$. Replacing $\mathcal{G}$ by its image in G , we may and shall assume that $\mathcal{G}$ is a subgroup scheme of G . Since the V_i are semi-simple $\mathcal{G}$ -modules, it follows that $\mathcal{G}$ is G -cr, for $G = \prod_j \mathrm{GL}(V_j)$. By [D, §6.2], we may also assume that $p > \dim(V_j) = h_{\mathrm{GL}(V_j)}$ for every j .

Hence, by working with the image of G , and $\mathcal{G}$, in $\mathrm{GL}(V)$, and applying 4.11, we conclude that V is semi-simple as a $\mathcal{G}$ -module. $\square$

Remark 4.14. If (V_i, q_i) is a non-degenerate quadratic space with $\dim V_i = 2d_i$ on which $\mathcal{G}$ acts by similitudes, then, by passing to a subgroup of index at most 2 and mapping to the group of similitudes rather than $\mathrm{GL}(V_i)$, one can replace the term $m_i(\dim V_i - m_i)$ by $m_i(\dim V_i - m_i - 1)$, when $m_i \leq 2d_i$.

Complete reducibility in the classical case. By 4.1, a subgroup scheme $\mathcal{G} \subset G$ is called cr if for every parabolic subgroup $P \subset G$ containing $\mathcal{G}$, there exists an opposite parabolic subgroup P' such that $\mathcal{G} \subset P \cap P'$. Suppose that $\mathrm{char}(k) \neq 2$ and let G be $\mathrm{SO}(V)$, or $\mathrm{Sp}(V)$ in any characteristic, relative to a non-degenerate symmetric or alternating bilinear form B on V . In this situation, the notion of cr can be interpreted as follows: $\mathcal{G}$ is cr in G if and only if for every $\mathcal{G}$ -submodule $W \subset V$ which is totally isotropic, there exists a totally isotropic $\mathcal{G}$ -submodule W' of the same dimension, such that the restriction of B to $W + W'$ is non-degenerate, cf. [S3, Example 3.3.3, page 206].

Lemma 4.15. *Let the subgroup scheme $\mathcal{G}$ of G be cr. Then the $\mathcal{G}$ -module V is semi-simple, and conversely.*

Proof. Let $W \subset V$ be a $\mathcal{G}$ -submodule. We need to produce a $\mathcal{G}$ -complement.

Consider $W_1 := W \cap W^\perp$. If $W_1 = (0)$, then $W \oplus W^\perp = V$ and we are done. So let $W_1 \neq (0)$. Then W_1 is a $\mathcal{G}$ -submodule which is totally isotropic; hence, by the cr property, we have a totally isotropic $\mathcal{G}$ -submodule W'_1 of the same dimension as W_1 , such that the form B is non-degenerate on $W_1 + W'_1$. In particular, $W_1 \cap W'_1 = (0)$. Since $W_1 \subset W^\perp$, we see that $W \subset W_1^\perp$.

Let $w \in W \cap W'_1 \subset W_1^\perp \cap W'_1$ and suppose $w \neq 0$. Since $w \in W'_1$, there exists $w' \in W_1$ such that $B(w, w') \neq 0$. On the other hand, since $w \in W_1^\perp$, $B(w, v) = 0$ for all $v \in W_1$, and in particular $B(w, w') = 0$, which contradicts the assumption that $w \neq 0$. Hence it follows that $W \cap W'_1 = (0)$.

Thus $W \subsetneq X = W \oplus W'_1 \subset V$ is a $\mathcal{G}$ -submodule. We proceed similarly and get $X_1 = X \cap X^\perp$ and a totally isotropic $\mathcal{G}$ -submodule X'_1 , of the same dimension, such that $X \oplus X'_1 \subset V$. If $X_1 = (0)$, then $V = X \oplus X^\perp$, so we get a $\mathcal{G}$ -decomposition of V as

$$W \oplus W'_1 \oplus \cdots,$$

that is, a $\mathcal{G}$ -complement of W in V .

Conversely, let V be semi-simple as a $\mathcal{G}$ -module. Let $W \subset V$ be a totally isotropic $\mathcal{G}$ -submodule of dimension d . Note that $d < \dim(V)/2$. We therefore have a $\mathcal{G}$ -submodule $Z \subset V$ such that $V = W \oplus Z$. The non-degenerate form B gives a $\mathcal{G}$ -equivariant isomorphism

$$\varphi : V \longrightarrow V^* = W^* \oplus Z^*.$$

Since W is totally isotropic, $\varphi(W) \cap W^* = (0)$. Hence $\varphi(W) \subset Z^*$.

Again, since W is totally isotropic, the restriction of B to Z is non-degenerate, and hence we get an isomorphism $\psi : Z^* \rightarrow Z$. Define

$$W' := \psi \circ \varphi(W).$$

Then it is easily seen that W' is of dimension d and is also a totally isotropic $\mathcal{G}$ -invariant submodule of V . Finally, B is non-degenerate on $W \oplus W'$. Hence $\mathcal{G}$ is cr in G . $\square$

5 Étale slices in positive characteristics

Let G act on an affine variety X and let x be a point of X whose orbit $G \cdot x$ is closed in X . We prove a schematic analogue of Luna's étale slice theorem (5.1, 5.6) for suitable bounds on the characteristic p . In order to make this precise, one needs to extend the notion of a slice in [BR, Definition 7.1] to a scheme-theoretic setting (see 5.6), and in the process one has to work with schematic stabilizers of closed orbits. The key issue then becomes the semisimplicity of the tangent space to x in X as a representation of the stabilizer of G at x . Here the structure theorem 2.5 becomes crucial, and the bounds on p are forced as a consequence. In [BR] the scheme-theoretic aspects are eschewed, forcing them to make the assumption that the orbit $G \cdot x$ is “separable”, i.e. the stabilizer G_x is reduced. The bounds on the characteristic therefore do not show up in [BR].

We begin this section with the following linear analogue of Luna's étale slice theorem in positive characteristics.

Theorem 5.1. *Let V be a G -module with low height, i.e. such that $p > \text{ht}_G(V)$. Let $v \in V$ be an element such that the orbit $G \cdot v$ is a closed orbit in V . Then there exists a G_v -invariant linear subspace S of V giving rise to a commutative diagram*

$$\begin{array}{ccc} G \times^{G_v} S & \xrightarrow{\phi} & V \\ f \downarrow & & \downarrow q \\ (G \times^{G_v} S) // G & \xrightarrow{\ell} & V // G \end{array} \quad (5.1.1)$$

and G -equivariant open subsets $U \subset (G \times^{G_v} S)$ containing the closed orbit $G \cdot v$, and an open subset U' of V containing v , for which (5.1.1) induces a cartesian diagram

$$\begin{array}{ccc} U & \xrightarrow{\phi} & U' \\ f \downarrow & & \downarrow q \\ U // G & \xrightarrow{\ell|_U} & U' // G \end{array} \quad (5.1.2)$$

such that the morphism $\ell|_U$ is étale.

Remark 5.2. The above theorem was stated in the note [MP], whose proof contained serious gaps, as was pointed out to the authors by Serre in private correspondence.

We note that G/G_v is constructed in [DG, III, Proposition 3.5.2]. It represents the quotient in the category of fppf sheaves. Furthermore, if $\pi_v : G \rightarrow V$, $g \mapsto g \cdot v$, the image $\text{im}(\pi_v)$, as a locally closed subscheme of V with its reduced scheme structure, can be identified with the scheme G/G_v . We call this locally closed subscheme the orbit $G \cdot v$, and have the identification $G/G_v \simeq G \cdot v$, see also [DG, Proposition and Definition 1.6, III, §3, page 325].

Proposition 5.3. *Let V be an arbitrary G -module. Let $v \in V$, and suppose that there exists a G_v -submodule S of the tangent space $T_v(V)$, such that*

$$T_v(V) = T_v(G \cdot v) \oplus S.$$

Let G act on $G \times S$ by $h \cdot (g, s) := (hg, s)$. Then the G -morphism $\Phi : G \times S \rightarrow V$, given by $(g, s) \mapsto g \cdot v + g \cdot s$, descends to a G -morphism

$$\phi : G \times^{G_v} S \longrightarrow V \tag{5.3.1}$$

which is étale at $(e, 0)$, where e is the identity of G .

Proof. To ensure that Φ descends to a morphism ϕ , we need to check, for every commutative k -algebra A , that Φ is constant on all $G_v(A)$ -orbits. For simplicity of exposition we suppress A .

Let $\alpha \in G_v$ act on $G \times S$ by

$$\alpha \cdot (g, s) = (g\alpha, \alpha^{-1}s).$$

Observe that

$$\Phi(\alpha \cdot (g, s)) = \Phi(g\alpha, \alpha^{-1}s) = g\alpha \cdot v + g\alpha\alpha^{-1} \cdot s = g \cdot v + g \cdot s,$$

since α fixes v . Therefore Φ is constant on the G_v -orbits. Since the action of G_v on $G \times S$ is scheme-theoretically free, Φ descends to a morphism $\phi : G \times^{G_v} S \rightarrow V$. Clearly the actions of G and G_v on $G \times S$ commute, and hence the descended morphism is also a G -morphism.

Observe that the quotient morphism $G \rightarrow G/G_v$ is a torsor for the group scheme G_v , locally trivial under the fppf topology. Since the action of G_v on S is linear, the associated fibre space

$$\pi : G \times^{G_v} S \longrightarrow G/G_v$$

is a locally free sheaf of rank $\dim(S)$. In particular, $G \times^{G_v} S$ is a smooth k -scheme of finite type. Observe further that, under the morphism ϕ , the zero section of the vector bundle $\pi : G \times^{G_v} S \rightarrow G/G_v$ canonically maps onto the orbit $G \cdot v \subset V$, while the fibre $\pi^{-1}(eG_v)$ of the identity coset $eG_v \in G/G_v$ maps isomorphically to the affine subspace $S + v \subset V$. Since $T_v(V) = T_v(G \cdot v) \oplus S$ by assumption, it follows that the differential $d\phi_z$, at $z = (e, 0)$, is an isomorphism.

We now apply [D, Lemme 2.9] to conclude that the morphism $\phi : G \times^{G_v} S \rightarrow V$ is étale at $z = (e, 0)$. $\square$

Proposition 5.4. *Let V be a G -module such that $p > \text{ht}_G(V)$, and let $v \in V$ be an element such that the G -orbit of v in V is closed. Then there exists a G_v -submodule $S \subset V$ such that*

$$V = T_v(G \cdot v) \oplus S$$

as a G_v -module. In particular, the consequences of 5.3 hold.

Proof. One knows that G_v^0 is the kernel of the natural map $G_v \rightarrow \pi_0(G_v)$ to the group of connected components, and hence is a normal subgroup of finite index. Further, we note that

$$|G_v/G_v^0| = |G_{v,\text{red}}/G_{v,\text{red}}^0|.$$

Note also that since $G_{v,\text{red}}$ is a saturated subgroup of G , by [S2, page 23, Property 3] the index $|G_{v,\text{red}}/G_{v,\text{red}}^0|$ is prime to p .

Since $G \cdot v$ is a closed orbit, by 4.12 and 2.5 we have an exact sequence

$$1 \longrightarrow G_{v,\text{red}}^0 \longrightarrow G_v^0 \longrightarrow \tau \longrightarrow 1, \tag{5.4.1}$$

where τ is a multiplicative group scheme.

Now V is semi-simple as a $G_{v,\text{red}}^0$ -module, $G_{v,\text{red}}^0$ being a saturated subgroup of G , and also as a τ -module, since τ is multiplicative and hence linearly reductive. Thus, by [D, Lemme 4.2], V is semi-simple as a G_v^0 -module and therefore as a G_v -module as well.

In particular, we have a G_v -supplement S for the G_v -invariant subspace $T_v(G \cdot v) \subset T_v(V) = V$, i.e. we have a G_v -decomposition $S \oplus T_v(G \cdot v)$ for V . $\square$

We recall the ‘‘Fundamental lemma of Luna’’, which holds in positive characteristics as well and which is essential to complete the proof of Theorem 5.1.

Lemma 5.5 ([GIT, page 152]). *Let $\bar{s} : X \rightarrow Y$ be a G -morphism of affine G -schemes. Let $F \subset X$ be a closed orbit such that:*

- (1) $\bar{s}$ is étale at some point of F ;
- (2) $\bar{s}(F)$ is closed in Y ;
- (3) $\bar{s}$ is injective on F ;
- (4) X is normal along F .

Then there are affine G -invariant open subsets $U \subset X$ and $U' \subset Y$, with $F \subset U$, such that

$$\bar{s} \parallel G : U \parallel G \longrightarrow U' \parallel G$$

is étale and

$$(\bar{s}, p_U) : U \longrightarrow U' \times_{U' \parallel G} U \parallel G$$

is an isomorphism, where $p_U : U \rightarrow U \parallel G$ is the quotient morphism.

Proof of 5.1. As a first step, we need to show that the categorical quotient $G \times^{G_v} S \parallel G$ exists as a scheme. As we have seen in 5.3, the action of G on $G \times S$, via $g \cdot (a, s) = (ga, s)$, and the twisted action of G_v on $G \times S$ commute, and hence

$$k[G \times^{G_v} S]^G = (k[G \times S]^{G_v})^G = (k[G \times S]^G)^{G_v} = k[S]^{G_v}. \quad (5.5.1)$$

Thus it is enough to show that $k[S]^{G_v}$ is finitely generated, and we would have

$$G \times^{G_v} S \parallel G \simeq S \parallel G_v. \quad (5.5.2)$$

As we have observed in the proof of 5.4, $G_v^0 \subset G_v$ is a normal subgroup of finite index. Thus, the finite generation of $k[S]^{G_v}$ is reduced to checking finite generation of $k[S]^{G_v^0}$. We may therefore replace G_v by G_v^0 in the proof. Further, since $G \cdot v$ is a closed orbit, $G_{v,\text{red}}$ is reductive. Therefore, by 4.12, we have a surjection

$$M \times G_{v,\text{red}} \longrightarrow G_v.$$

Since M in G_v is central,

$$k[S]^{G_v} = (k[S]^{G_{v,\text{red}}})^M.$$

Again, since $G_{v,\text{red}}$ is reductive, the ring of invariants $k[S]^{G_{v,\text{red}}}$ is a finitely generated k -algebra. Since M is a group of multiplicative type over an algebraically closed field, it is a product of $\mathbf{G}_m$'s and a finite group scheme. Therefore, by [GIT, Theorem 1.1] and [AV, page 113], the ring of invariants $(k[S]^{G_{v,\text{red}}})^M$ is also finitely generated. This proves the finite generation of $k[S]^{G_v}$.

The commutativity of the diagram (5.1.1) now follows by the property of categorical quotients by the group G .

Now we check the conditions of Lemma 5.5. We consider the given closed G -orbit

$$F = G \times^{G_v} \{v\} \subset G \times^{G_v} S.$$

Then we need to check that $\phi(F)$ is a closed orbit in V . In fact, $\phi(F)$ is precisely the closed orbit $G \cdot \phi(v) = G \cdot v \subset V$. Thus we have verified conditions (2) and (3) of Lemma 5.5. Condition (1) is precisely the content of 5.4, and (4) holds since $G \times^{G_v} S$ has been seen to be smooth.

The isomorphism $(\bar{s}, p_U)$ shows that the diagram (5.1.2) is cartesian. This completes the proof of 5.1.

Corollary 5.6. *Let X be an affine G -scheme embeddable as a closed G -subscheme in a low height G -module. Let $x \in X$ be such that the orbit $G \cdot x \subset X$ is closed. Then there exists a locally closed G_x -invariant subscheme, a “slice” X_1 , of X , with $x \in X_1 \subset X$, such that the conclusions of 5.1 hold for the G -morphism $G \times^{G_x} X_1 \rightarrow X$.*

Proof. This follows from 5.1, which corresponds to [BR, Proposition 7.4], exactly as in [BR, Proposition 7.6]. $\square$

The following consequence of the slice theorem has many applications, so we state it here without proof.

Corollary 5.7 (see [BR, Proposition 8.5, page 312]). *Let F be an affine G -subvariety of $\mathbb{P}(V)$, with V a G -module with low height, and suppose that F contains a unique closed orbit F^{cl} . Then there exists a G -retract*

$$F \longrightarrow F^{\text{cl}}.$$

Let V be a finite dimensional G -module and let

$$\text{ht}_G(\bigwedge(V)) := \max_i \{\text{ht}_G(\bigwedge^i V)\}. \quad (5.7.1)$$

We then have the following application to the theory of semistable principal bundles in positive characteristics.

Theorem 5.8. *Let E be a stable G -bundle with G semi-simple and $\rho_V : G \rightarrow \text{SL}(V)$ be a representation such that $p > \text{ht}_G(\bigwedge(V))$. Then the associated bundle $E(V)$ is polystable of degree 0.*

Proof. The proof follows [BP, Theorem 9.11] verbatim, where the notion of “separable index” is replaced by the Dynkin height, the key ingredient being Corollary 5.7. $\square$

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6 Added in proof

In Section 2.2, after making suitable assumptions on G , we refer to Serre in (2.2.2) for the construction of the exponential map

$$\exp : \mathfrak{g}_{\text{nilp}} \xrightarrow{\sim} G^{\text{u}}.$$

The proof that Serre gives in [S2, Theorem 3, page 21] is only a sketch. At his suggestion, we complete it. We work in principle over an algebraically closed field, but phrase the argument so that it remains valid over more general bases.

Let T be a maximal torus, B a Borel subgroup containing T , and U its unipotent radical. We assume $p \geq h$. By this assumption on the characteristic, the Campbell-Hausdorff group law $\circ$ on $\text{Lie}(U)$ is well-defined. The first step is the construction of a B -equivariant isomorphism

$$\exp : (\text{Lie}(U), \circ) \xrightarrow{\sim} U. \quad (6.0.1)$$

For this step we give complete proofs. The second step is to glue these isomorphisms into

$$\exp : \mathfrak{g}_{\text{nilp}} \xrightarrow{\sim} G^u. \quad (6.0.2)$$

This requires the additional assumption that the simply connected covering $\tilde{G}$ of the derived group G' is étale over G' . Here we will mainly give references.

6.1

Let R^+ be the set of positive roots: the set of roots of T acting on $\text{Lie}(U)$. For $\alpha \in R^+$, let U_α be the corresponding root subgroup, and let x_α be an isomorphism from U_α to $\mathbf{G}_a$.

Fix a total order on R^+ . The corresponding product map from $\prod U_\alpha$ to U is an isomorphism of schemes: it is a morphism between smooth connected schemes which is étale at the origin, bijective by [Bible, 13-05], and one applies Zariski's main theorem as in [SGA3, XXII, 4.1], or see [Bo2, IV, 14.5].

This isomorphism identifies U with the affine space with coordinates $(x_\alpha)_{\alpha \in R^+}$. In these coordinates, the action of T on U by conjugation is given by

$$t \in T \quad \text{acts by} \quad (x_\alpha) \longmapsto (\alpha(t)x_\alpha).$$

This action respects the group law. It follows that the α -coordinate of $x \cdot y$ is a linear combination of the $x_\beta^i y_\gamma^j$ for $i\beta + j\gamma = \alpha$.

We grade the polynomial algebra $\mathcal{O}(U)$ by defining the degree of x_α to be the height of α . If the expression of α as a linear combination of simple roots is $\sum n_i \alpha_i$, the height of α is $\sum n_i$. The corresponding increasing filtration Fil of $\mathcal{O}(U)$ is invariant by T -conjugation as well as by the action on the right and left by U -translations. It is hence invariant by B -conjugation.

The assumption $p \geq h$ is equivalent to $\mathcal{O}(U)$ being generated by Fil_{p-1} . The definition of the exponential (6.0.1) is a special case of the following construction.

6.2

Let U be an algebraic group and let Fil be an increasing filtration of its affine algebra. Assume that $\mathcal{O}(U)$ is a polynomial algebra $k[(x_\alpha)_{\alpha \in A}]$, and that for some integers $d(\alpha) > 0$, Fil is the increasing filtration deduced from the grading for which x_α is homogeneous of degree $d(\alpha)$. Assume that the α -coordinate of $x \cdot y$ is a polynomial of degree $\leq d(\alpha)$ in the weighted coordinates of x and y . This is equivalent to the compatibility

$$\Delta(\text{Fil}_i) \subset \sum_{j+k=i} \text{Fil}_j \otimes \text{Fil}_k \quad (6.2.1)$$

of the filtration Fil with the coproduct.

We will view $\text{Lie}(U)$ as the Lie algebra of the left invariant vector fields on U . Let D_α be the element of $\text{Lie}(U)$ which is ∂_α at e . If $[\varepsilon]_\alpha$ is the $k[\varepsilon]/(\varepsilon^2)$ -valued point of U with coordinate $x_\alpha = \varepsilon$, and $x_\beta = 0$ for $\beta \neq \alpha$, then $D_\alpha f$ is the coefficient of ε in $f(x \cdot [\varepsilon]_\alpha)$. It follows that $D_\alpha(x_\beta)$ is of degree $\leq d(\beta) - d(\alpha)$, and consequently

$$D_\alpha \text{Fil}_i \subset \text{Fil}_{i-d(\alpha)}. \quad (6.2.2)$$

Thus D_α is of descending filtration $d(\alpha)$.

We now assume that $\mathcal{O}(U)$ is generated by Fil_{p-1} , i.e. that all $d(\alpha)$ are less than p . Then any Lie polynomial of degree $\geq p$ in the derivations D_i in $\text{Lie}(U)$ vanishes on the x_α , and hence vanishes identically. It follows that the Campbell–Hausdorff group law $\circ$ on $\text{Lie}(U)$ is defined.

Definition 6.3. The exponential map $\exp$ from $\text{Lie}(U)$ to U is the map having as coordinates the truncated exponential

$$\exp(D)_\alpha := \sum_{n < p} \left(\frac{D^n}{n!}(x_\alpha) \text{ at } e \right). \quad (6.3.1)$$

Lemma 6.4. For $f \in \text{Fil}_{p-1}$, one has

$$f(\exp(D)) = \sum_{n < p} \left(\frac{D^n}{n!}(f) \text{ at } e \right). \quad (6.4.1)$$

Proof. This holds by the definition of $\exp$ when f is a coordinate function x_α . It therefore suffices to check that, for $g \in \text{Fil}_a$ and $h \in \text{Fil}_b$ with $a + b < p$, one has

$$\sum \frac{D^n}{n!}(gh) = \sum \frac{D^n}{n!}(g) \sum \frac{D^n}{n!}(h), \quad (6.4.2)$$

the sums being truncated at $n < p$. Both sides can be identified with sums $\sum \frac{D^n}{n!}(g) \frac{D^m}{m!}(h)$, where on the left we have $n + m < p$, and on the right $n < p$ and $m < p$. If $n + m \geq p$, then either $n > a$ or $m > b$, in which case $D^n(g) = 0$, respectively $D^m(h) = 0$. The terms missing on the left therefore vanish. $\square$

Lemma 6.5. The morphism $\exp$ is a group homomorphism

$$\exp : (\text{Lie}(U), \circ) \longrightarrow U. \quad (6.5.1)$$

Proof. If $f \in \text{Fil}_{p-1}$, then so are its left-translates. Since the derivations D in $\text{Lie}(U)$ are left invariant, (6.4.1) applied to the left-translate of f gives

$$f(x \cdot \exp(D)) = \sum_{n < p} \left(\frac{D^n}{n!}(f) \text{ at } x \right), \quad (6.5.2)$$

hence

$$f(\exp(D') \cdot \exp(D)) = \sum_{n, m < p} \left(\frac{(D')^m}{m!} \frac{D^n}{n!}(f) \text{ at } e \right). \quad (6.5.3)$$

In the completed free associative algebra $\mathbb{Q}\langle\langle X, Y \rangle\rangle$, if $H_d(X, Y)$ is the Lie polynomial which is the degree d component of the Campbell–Hausdorff group law, one has

$$\exp \left(\sum H_d(X, Y) \right) = \exp(X) \cdot \exp(Y). \quad (6.5.4)$$

Modulo terms of degree $\geq p$, one still has

$$\sum_{n \leq p} \frac{\left(\sum_{d < p} H_d(X, Y) \right)^n}{n!} \equiv \sum_{n < p} \frac{X^n}{n!} \sum_{n < p} \frac{Y^n}{n!}, \quad (6.5.5)$$

and this continues to hold in $\mathbb{Z}[1/(p-1)]\langle\langle X, Y \rangle\rangle$. Substituting D' and D for X and Y , we get

$$f(\exp(D') \cdot \exp(D)) = f(\exp(D' \circ D)) \quad (6.5.6)$$

for $f \in \text{Fil}_{p-1}$. Hence $\exp(D') \cdot \exp(D) = \exp(D' \circ D)$.

Define $\Delta_a f$ to be $f(xa) - f(x)$. For $f \in \text{Fil}_{p-1}$, (6.5.2) reads

$$\Delta_{\exp(D)} f = \sum_{0 \leq n < p} \frac{D^n}{n!}(f). \quad (6.5.7)$$

In $\mathbb{Q}[[X]]$, one has $\log \exp(X) = X$. Modulo monomials of degree $\geq p$, one still has

$$\sum_{0 < n < p} (-1)^{n+1} \left(\sum_{0 < m < p} \frac{X^m}{m!} \right)^n / n \equiv X, \quad (6.5.8)$$

and this continues to hold in $\mathbb{Z}[1/(p-1)!][[X]]$. Substituting D for X in (6.5.8), we obtain from (6.5.7) that, for $f \in \text{Fil}_{p-1}$,

$$Df = \sum_{0 < n < p} (-1)^{n+1} \Delta_{\exp(D)}^n f / n. \quad (6.5.9)$$

This gives a formula for a left inverse $a \mapsto \log a$ of $\exp : \text{Lie}(U) \rightarrow U$; viewed as a derivation,

$$\log a = \sum_{0 < n < p} (-1)^{n+1} \Delta_a^n / n \quad (6.5.10)$$

when applied to $f \in \text{Fil}_{p-1}$. This shows that the exponential is injective and hence, being an étale homomorphism of connected groups, it is an isomorphism.

In the case of the unipotent radical U of a Borel subgroup, the given construction of the exponential map, depending only on the filtration Fil of $\mathcal{O}(U)$, is invariant by B -conjugation. $\square$

Remark 6.6. (i) Let $I \subset R^+$ be the set of simple roots, and for $J \subset I$, let P_J be the corresponding parabolic subgroup and U_J its unipotent radical. Let $R_J^+ \subset R^+$ be the set of $\alpha \in R^+$ which are linear combinations of simple roots in J . For $\alpha = \sum n_i \alpha_i$ in $R^+ - R_J^+$, define $d_J(\alpha) = \sum_{i \notin J} n_i$. Then, if all $d_J(\alpha)$ are $< p$, the construction we have given of the exponential map gives a similar map

$$\exp : \text{Lie}(U_J) \longrightarrow U_J. \quad (6.6.1)$$

It suffices to replace d by d_J in the earlier proofs.

(ii) More generally, let $\Gamma \subset R^+$ be a closed set of positive roots, and let U_Γ be the corresponding subgroup of U . For any fixed total order on Γ , the corresponding product map

$$\prod_{\alpha \in \Gamma} U_\alpha \longrightarrow U_\Gamma \quad (6.6.2)$$

is an isomorphism of schemes. For $\alpha \in \Gamma$, define $d_\Gamma(\alpha)$ to be the largest number of elements of Γ of which α is the sum, with repetitions allowed.

In [Sei], Seitz states that when all $d_\Gamma(\alpha)$ are $< p$, an exponential map from $\text{Lie}(U_\Gamma)$ to U_Γ is defined. He writes, before [Sei, Proposition 5.1, page 483], that the argument in Section 4 of [S1] yields this result. The reference is mistaken and should be replaced by Section 2.1 of [S4]. The argument given in the preceding paragraphs, with d replaced by d_Γ , also yields the result.

6.7

Let $\mathcal{B}$ be the variety of Borel subgroups of G . We have fiber spaces X and Y over $\mathcal{B}$, where the fibers at a Borel subgroup B in $\mathcal{B}$ are respectively the unipotent radical U and its Lie algebra. The given construction of the exponential map works over any base on which the primes $\ell < h$ are invertible. Over $\mathcal{B}$, we therefore obtain an isomorphism $\exp : Y \xrightarrow{\sim} X$. The natural map from X to G , respectively from Y to $\text{Lie}(G)$, is proper, being the restriction to a closed subset of $\mathcal{B} \times G$, respectively $\mathcal{B} \times \text{Lie}(G)$, of the second projection.

The reduced subscheme $\mathfrak{g}_{\text{nilp}}$, respectively G^u , of $\text{Lie}(G)$, respectively G , with points the nilpotent, respectively unipotent, elements, is the schematic image of Y in $\text{Lie}(G)$, respectively

of X in G . We want to complete the following G -equivariant diagram in the solid arrows by an isomorphism at the dotted arrow:

$$\begin{array}{ccc} Y & \xrightarrow{\exp} & X \\ \downarrow & & \downarrow \\ \mathfrak{g}_{\text{nilp}} & \dashrightarrow & G^u \end{array} \quad (6.7.1)$$

Replacing G by its derived group G' does not change the diagram, nor does replacing a semi-simple G by its simply connected covering $\tilde{G}$, provided $\tilde{G}$ is étale over G . Under our assumptions this reduces us to the case where G is semi-simple and simply connected. Here are references for the existence of the dotted arrow isomorphism in (6.7.1).

(A) By [St, 6.11, 8.1], G^u is normal with a smooth dense orbit whose complement is of codimension ≥ 2 . This holds in any characteristic for any simply connected G . Normality results from G^u being the complete intersection in G defined by the equations $\chi(g) = \chi(e)$ for χ the characters of the fundamental representations, and being smooth outside codimension 2.

(B) The same holds for $\mathfrak{g}_{\text{nilp}}$ when G is simply connected and p good, because $\mathfrak{g}_{\text{nilp}}$ is then equivariantly isomorphic to G^u . A nice proof is in [BR, Corollary 9.3.4]. The isomorphism uses a representation (V, ρ) such that the pairing $\text{Tr}(\rho(X)\rho(Y))$ on $\text{Lie}(G)$ is non-degenerate. It is then defined as induced by the map $g \mapsto \rho(g) - 1$ from G to $\text{End}(V)$, composed with the orthogonal projection of $\text{End}(V)$, endowed with the symmetric bilinear form $\text{Tr}(XY)$, to $\text{Lie}(G) \subset \text{End}(V)$. This does not work for $\text{SL}(n)$, for which one uses $X \mapsto 1 + X$.

(C) Over the dense orbits of $\mathfrak{g}_{\text{nilp}}$ and G^u , (6.7.1) is a system of isomorphisms, and the composite isomorphism extends across codimension 2 by normality.

Appendix. Coxeter number and root systems, by Zhiwei Yun

Proposition A1. *Let R be an irreducible root system associated to a simple group G with Coxeter number h , and let X be the lattice spanned by R . Let $\varphi : X \rightarrow \mathbb{R}/\mathbb{Z}$ be a homomorphism. Then there exists a basis B for R such that if $\alpha \in R$ satisfies $\varphi(\alpha) \in (0, 1/h) \bmod \mathbb{Z}$, then α is positive with respect to B .*

Proof. Pick any basis $B_0 = \{\alpha_1, \dots, \alpha_r\}$ of R . We lift φ to a linear map $y : X \rightarrow \mathbb{R}$, which is viewed as a point in $V = \text{Hom}(X, \mathbb{R})$.

Let W be the Weyl group of R . Then there is an action of the extended Weyl group $\text{Hom}(X, \mathbb{Z}) \rtimes W$ on V , and the validity of the statement for the linear map y is invariant for this action. Therefore we may assume that y lies in the fundamental alcove attached to B_0 , i.e.

$$\alpha_j(y) \geq 0, \quad 1 \leq j \leq r, \quad (\text{A1.1})$$

$$\theta(y) \leq 1, \quad (\text{A1.2})$$

where θ is the highest root with respect to B_0 . Then, for all $\alpha \in R$,

$$|\alpha(y)| \leq 1. \quad (\text{A1.3})$$

We claim that there exists a basis B of R such that, for any $\beta \in R$ with

$$\beta(y) \in (-1, -1 + 1/h) \cup (0, 1/h), \quad (\text{A1.4})$$

β is positive with respect to B .

Write $\theta = \sum_i n_i \alpha_i$, and let $\alpha_0 = 1 - \theta$ and $n_0 = 1$, as usual. Then we have the equality $\sum_{i=0}^r n_i \alpha_i = 1$, viewed as affine functions on V , and we have

$$h = \sum_{i=0}^r n_i. \quad (\text{A1.5})$$

Therefore, for some $0 \leq i \leq r$, we have $\alpha_i(y) \geq 1/h$.

If $i = 0$, then $\theta(y) \leq 1 - 1/h$, and in this case we can take $B = B_0$. In fact, for any negative root $-\beta$ with respect to B_0 , we have $0 \geq -\beta(y) \geq -1 + 1/h$, so that $-\beta$ cannot satisfy condition (A1.4).

If $i > 0$, let v_i be the vertex of the fundamental alcove opposite to the root hyperplane α_i . Thus $\alpha_j(v_i) = 0$ if $j \neq i$, and $\alpha_i(v_i) = 1/n_i$. We take a basis B of R such that the vector $y - v_i$ is dominant, i.e. for any $\beta \in R$ positive for B , one has $\beta(y) \geq \beta(v_i)$.

Such a basis B satisfies the requirements of the claim. In fact, let $\beta \in R$ be negative with respect to B . Write $\beta = \sum_{j=1}^r m_j \alpha_j$, with $-n_j \leq m_j \leq n_j$. Then $\beta(v_i) = m_i/n_i$. On the other hand, $\beta(y) = \sum_{j=1}^r m_j \alpha_j(y)$. Since β is negative for B , we have

$$\beta(y) \leq \beta(v_i) = m_i/n_i. \quad (\text{A1.6})$$

We have two cases.

- If $\beta > 0$ with respect to B_0 , then $\beta(y) \geq 0$ and $0 \leq m_j \leq n_j$ for all j . If $m_i = 0$, then (A1.6) implies that $\beta(y) = 0$, and so β fails to satisfy (A1.4). If $m_i \geq 1$, then $\beta(y) \geq \alpha_i(y) \geq 1/h$, again not satisfying (A1.4).
- If $\beta < 0$ with respect to B_0 , then $-1 \leq \beta(y) \leq 0$ and $-n_j \leq m_j \leq 0$ for all j . If $m_i = -n_i$, then $\beta(v_i) = -1$, which forces $\beta(y) = -1$ by (A1.6). If $m_i \geq -n_i + 1$, then, comparing $|\beta(y)|$ with $\theta(y)$, it misses at least one copy of $\alpha_i(y)$; therefore $|\beta(y)| \leq \theta(y) - \alpha_i(y) \leq 1 - 1/h$, and again $\beta(y)$ does not satisfy (A1.4).

This proves the claim and the proposition. □

References

- [BP] V. Balaji and A. J. Parameswaran, *Tensor product theorem for Hitchin pairs – An algebraic approach*, Ann. Inst. Fourier (Grenoble) 61 (2011), no. 6, 2361–2403. MR-2976315.
- [BR] P. Bardsley and R. W. Richardson, *Étale slices for algebraic transformation groups in characteristic p* , Proc. London Math. Soc. (3) 51 (1985), no. 2, 295–317. MR-0794118.
- [BMR1] M. Bate, B. Martin, and G. Röhrle, *A geometric approach to complete reducibility*, Invent. Math. 161 (2005), no. 1, 177–218. MR-2178661.
- [BMR2] M. Bate, B. Martin, and G. Röhrle, *Complete reducibility and commuting subgroups*, J. Reine Angew. Math. 621 (2008), 213–235. MR-2431255.
- [Bible] C. Chevalley, *Séminaire 1956–1958, Classification des groupes de Lie algébriques*, multigraphié, Paris, Secrétariat mathématique.
- [Bo] A. Borel, *On affine algebraic homogeneous spaces*, Arch. Math. 45 (1985), no. 1, 74–78. MR-0799451.
- [Bo2] A. Borel, *Linear Algebraic Groups*, 2nd ed., Springer, GTM, 1991. MR-404273.
- [BT] A. Borel and J. Tits, *Éléments unipotents et sous-groupes paraboliques de groupes réductifs. I*, Invent. Math. 12 (1971), 95–104. MR-0294349.
- [CGP] B. Conrad, O. Gabber, and G. Prasad, *Pseudo-reductive groups*, second edition, New Mathematical Monographs, vol. 26, Cambridge University Press, Cambridge, 2015. MR-3362817.

- [D] P. Deligne, *Semi-simplicité de produits tensoriels en caractéristique p* , Invent. Math. 197 (2014), no. 3, 587–611. MR-3251830.
- [DG] M. Demazure and P. Gabriel, *Groupes algébriques. Tome I: Géométrie algébrique, généralités, groupes commutatifs*, Masson; North-Holland, 1970. MR-0302656.
- [Dy] E. B. Dynkin, *Maximal subgroups of the classical groups*, Trudy Moskov. Mat. Obshch. 1 (1952), 39–166; English translation in Amer. Math. Soc. Transl. (2) 6 (1957), 245–378. MR-0049903.
- [H] J. E. Humphreys, *Linear algebraic groups*, Graduate Texts in Mathematics, vol. 21, Springer-Verlag, New York–Heidelberg, 1975. MR-0396773.
- [IMP] S. Ilangovan, V. B. Mehta, and A. J. Parameswaran, *Semistability and semisimplicity in representations of low height in positive characteristic*, in *A tribute to C. S. Seshadri*, Trends Math., Birkhäuser, Basel, 2003, pp. 271–282. MR-2017588.
- [Lu] D. Luna, *Slices étales*, in *Sur les groupes algébriques*, Bull. Soc. Math. France, Mémoire 33, Soc. Math. France, Paris, 1973, pp. 81–105. MR-0342523.
- [Mc1] G. J. McNinch, *Abelian unipotent subgroups of reductive groups*, J. Pure Appl. Algebra 167 (2002), no. 2–3, 269–300. MR-1874545.
- [Mc2] G. J. McNinch, *Optimal $SL(2)$ -homomorphisms*, Comment. Math. Helv. 80 (2005), no. 2, 391–426. MR-2142248.
- [MP] V. B. Mehta and A. J. Parameswaran, *Geometry of low height representations*, in *Algebra, Arithmetic and Geometry*, Part I, II, Tata Inst. Fund. Res. Stud. Math., vol. 16, Bombay, 2002, pp. 417–426. MR-1940675.
- [AV] D. Mumford, *Abelian varieties*, with appendices by C. P. Ramanujam and Yuri Manin, corrected reprint of the second (1974) edition, Tata Inst. Fund. Res. Stud. Math., vol. 5, Hindustan Book Agency, New Delhi, 2008. MR-2514037.
- [GIT] D. Mumford, J. Fogarty, and F. Kirwan, *Geometric invariant theory*, third edition, Ergebnisse der Mathematik und ihrer Grenzgebiete (2), vol. 34, Springer-Verlag, Berlin, 1994. MR-1304906.
- [Sei] G. M. Seitz, *Unipotent elements, tilting modules, and saturation*, Invent. Math. 141 (2000), no. 3, 467–502. MR-1779618.
- [S1] J.-P. Serre, *Sur la semi-simplicité des produits tensoriels de représentations de groupes*, Invent. Math. 116 (1994), no. 1–3, 513–530. MR-1253203.
- [S2] J.-P. Serre, *Moursund Lectures 1998 at the University of Oregon Mathematics Department*, 1998, arXiv:math/0305257.
- [S3] J.-P. Serre, *Complète réductibilité*, Séminaire Bourbaki, vol. 2003/2004, Astérisque 299 (2005), exp. no. 932, pp. 195–217. MR-2167207.
- [S4] J.-P. Serre, *Exemples de plongements des groupes $PSL_2(\mathbb{F}_p)$ dans des groupes de Lie simples*, Invent. Math. 124 (1996), 525–562. MR-1369427.
- [SGA3] *Séminaire de Géométrie Algébrique du Bois Marie 1962/64 (SGA 3)*, dirigé par M. Demazure et A. Grothendieck, Schémas en Groupes, Lecture Notes in Mathematics, vols. 151–153, Springer, Berlin, 1970. MR-0274458.

- [St] R. Steinberg, *Regular elements of semisimple algebraic groups*, Publ. Math. IHES 25 (1965), 49–80 (= Collected Papers no. 20). MR-314788.
- [W1] D. J. Winter, *On the toral structure of Lie p -algebras*, Acta Math. 123 (1969), 69–81. MR-0251095.
- [W2] D. J. Winter, *Abstract Lie algebras*, M.I.T. Press, Cambridge, Mass.–London, 1972; Dover Publications, 2008. MR-0332905.